



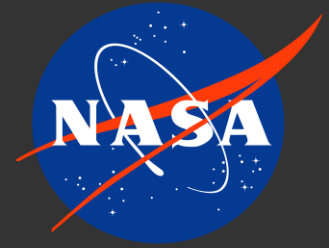
VANDERBILT AEROSPACE DESIGN LABORATORY

2026 NASA Student Launch Initiative

Preliminary Design Review

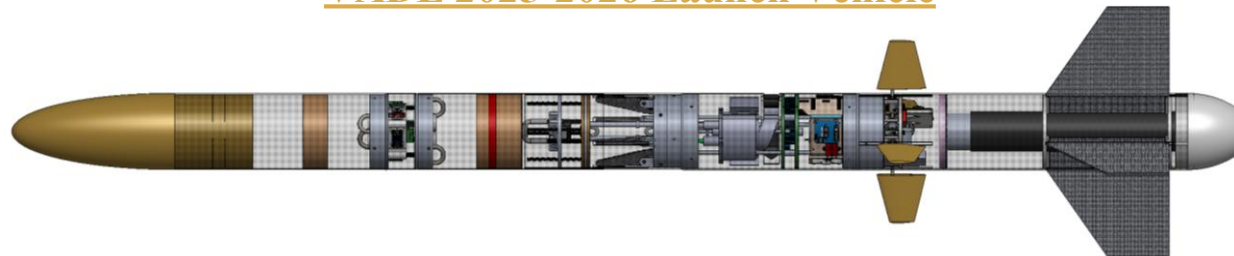
Mission Statement

NASA 2025 Student Launch Initiative



The VADL 2025-2026 launch vehicle will achieve a controlled apogee of **4,200 ft AGL** with the use of an **Apogee Control System**. The rocket will recover via **dual-deployment**—drogue at apogee and main at ≥ 500 ft AGL—so all sections land **safely** and are ready to re-launch without repair. After touchdown, the tail-mounted **R.O.L.L** payload will autonomously acquire ≥ 50 **mL** of soil with a rotating shovel and analyze the retained sample for required metrics (e.g., **nitrate-nitrogen, pH, electrical conductivity**). Soil collection and sensing will be completed in less than **15 minutes**, with results recorded for post-flight reporting.

VADL 2025-2026 Launch Vehicle



VANDERBILT
UNIVERSITY



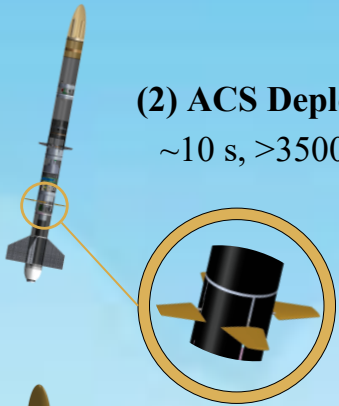
VANDERBILT AEROSPACE
DESIGN LABORATORY

(3) Apogee, Drogue Deployment



(2) ACS Deployment

~10 s, >3500' AGL



(4) Main Deployment

60 s, 550' AGL

(5) Landing

102 s

Mission Overview



(6) Parachute Detachment

113 s

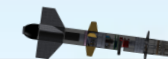
(7) Payload Activation

t = 115 s



(1) Launch

0 s



Presenting Team Introduction



Jack Liedel
President



Ashley Fu
Electrical Lead



Jamie Danis
Vehicle Lead



Thor Hammer
Electrical Eng.



Evan Levine
Operation Lead



Evan Ticknor
Payload Eng.



Ben Racapé
Vice President



Sean Confoy
Payload Lead



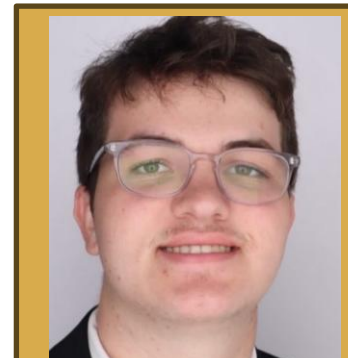
Joel Berinstein
Electrical Eng.



Catherine Knox
Payload Eng.



Sadie Pullen
Electrical Eng.



Caleb Strout
Simulation Eng.

Leading Launch Vehicle Design



Launch Vehicle Overview and Dimensions

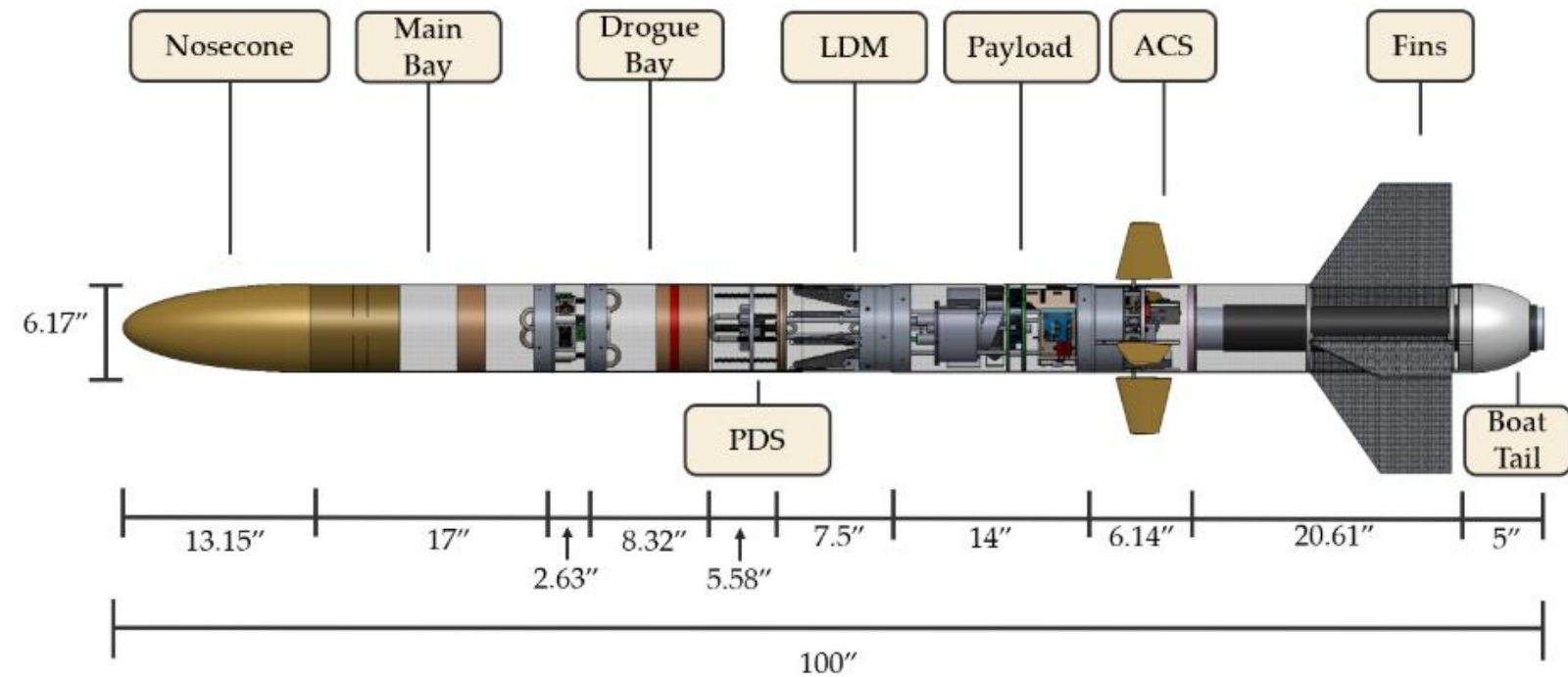


Figure 1: Fullscale Vehicle Dimensions
AR = 16.2:1

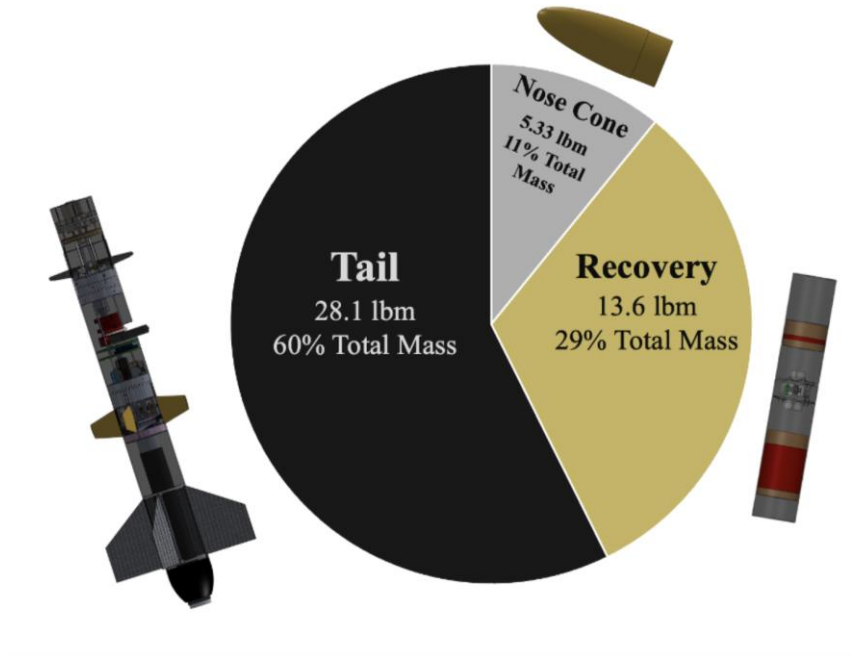
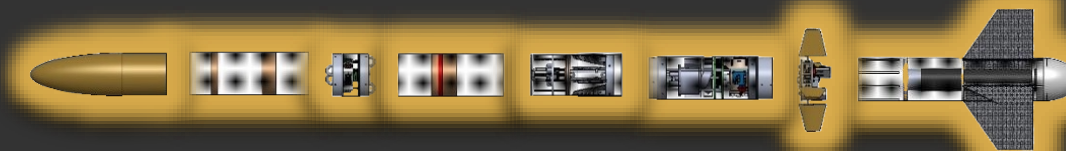


Figure 2: Fullscale Vehicle Section Weights





Launch Vehicle Materials and Alternatives

Airframe Material

*Commercially Available G12
Fiberglass Tubes*

- Highly workable
- Consistent tolerances



Figure 1: Standard 6 inch ID
Fiberglass Tube Used for
Airframe Material

Bulkplate Material

6061-T6 Aluminum

- High strength to weight ratio
- Low cost
- Reusability and adaptability

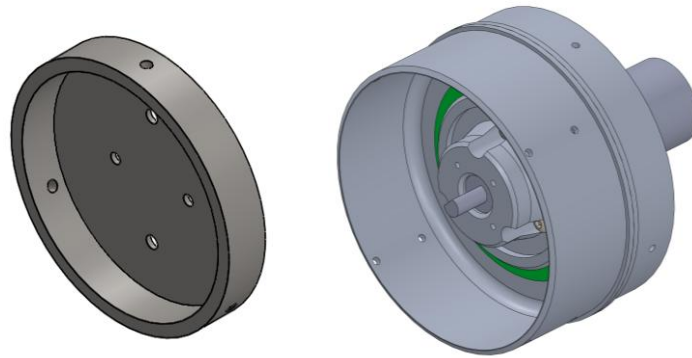


Figure 2: Standard Bulkplate
and Rotating Bulkplates on the
Left and Right Respectively

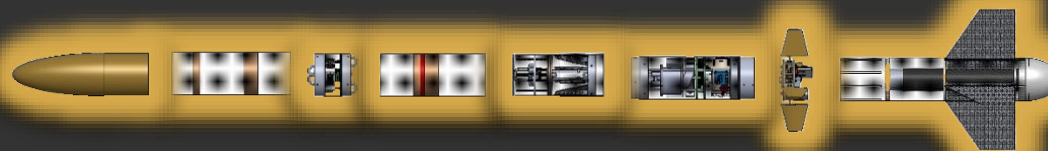
Fin and Leg Material

Dragon Plate Carbon Fiber

- High shear & flexural strength



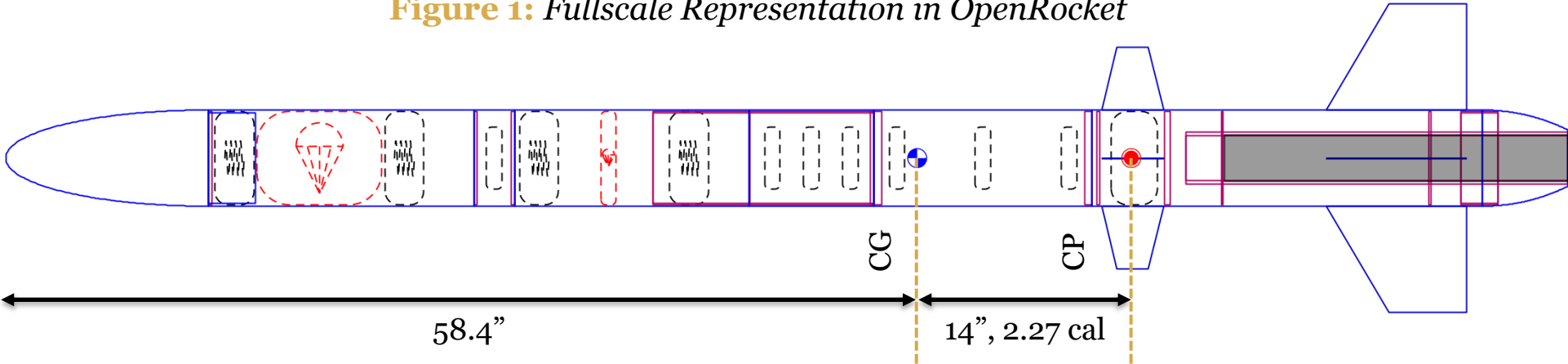
Figure 3: Standard Carbon
Fiber without and with Resin
Reinforcement on the Top and
Bottom Respectively



Launch Vehicle Stability

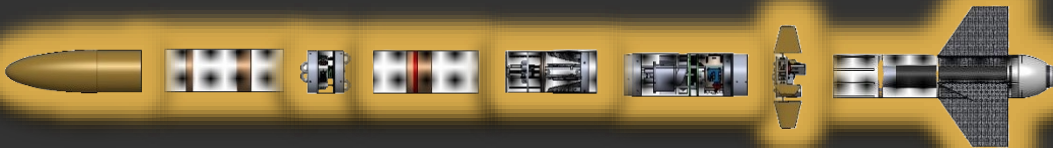


Figure 1: Fullscale Representation in OpenRocket



Flight Phase	CG (in)	CP (in)	SSM (cal)
On Pad	58.4	72.4	2.27
Launch Rail Exit	58.1	72.4	2.32
Post MECO	55.8	72.4	2.70

Table 1: Static Stability Margin (SSM) During Flight





Nose Cone Selection and Ballast System

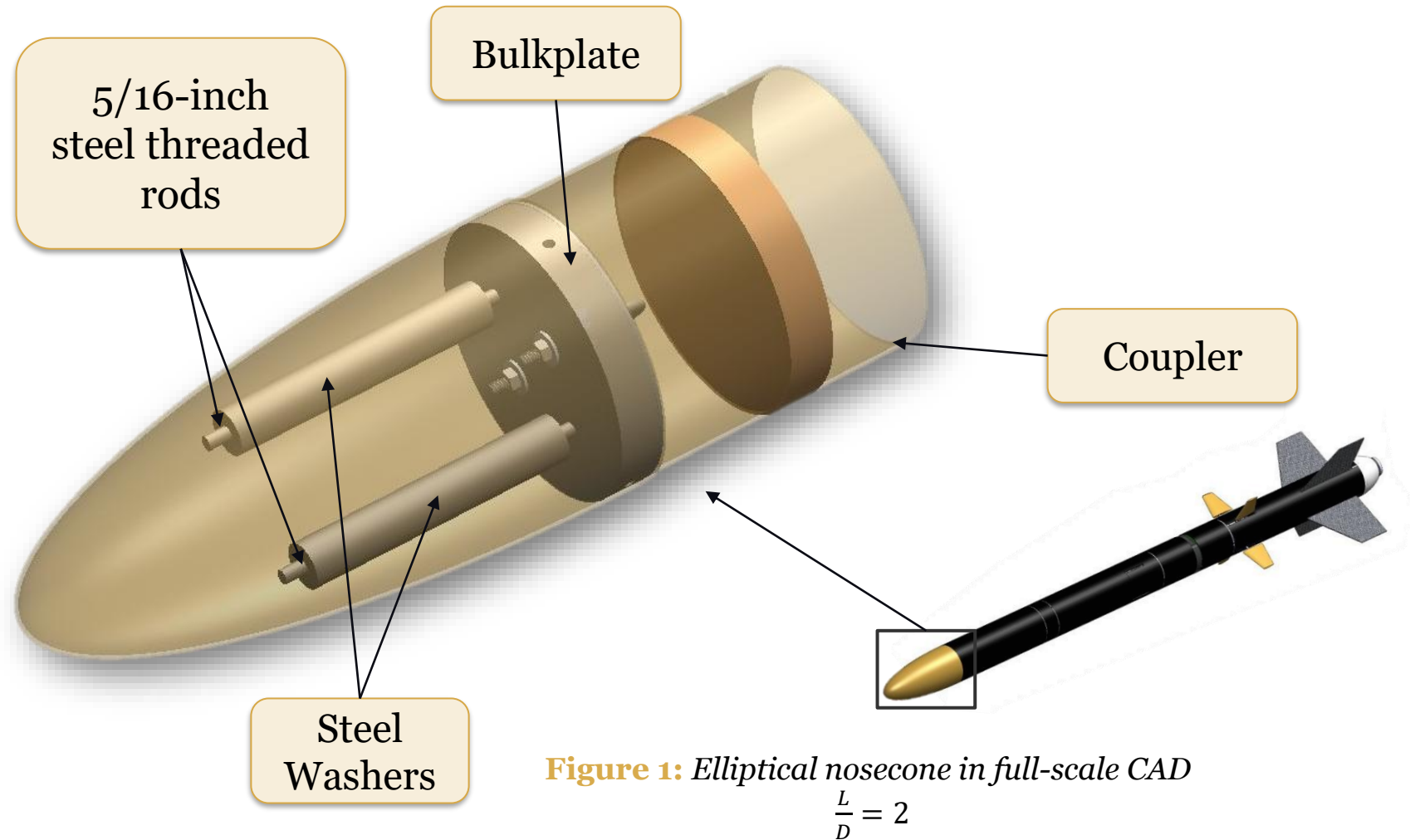


Figure 1: Elliptical nosecone in full-scale CAD

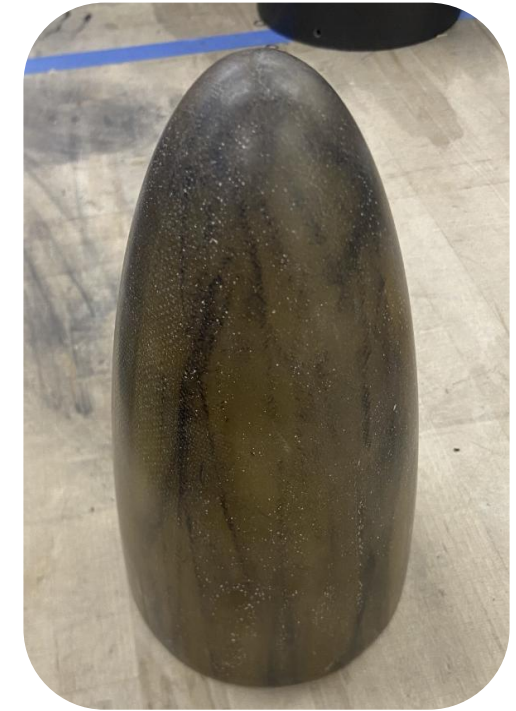


Figure 2: Nosecone Fabrication





Vehicle: Recovery Section

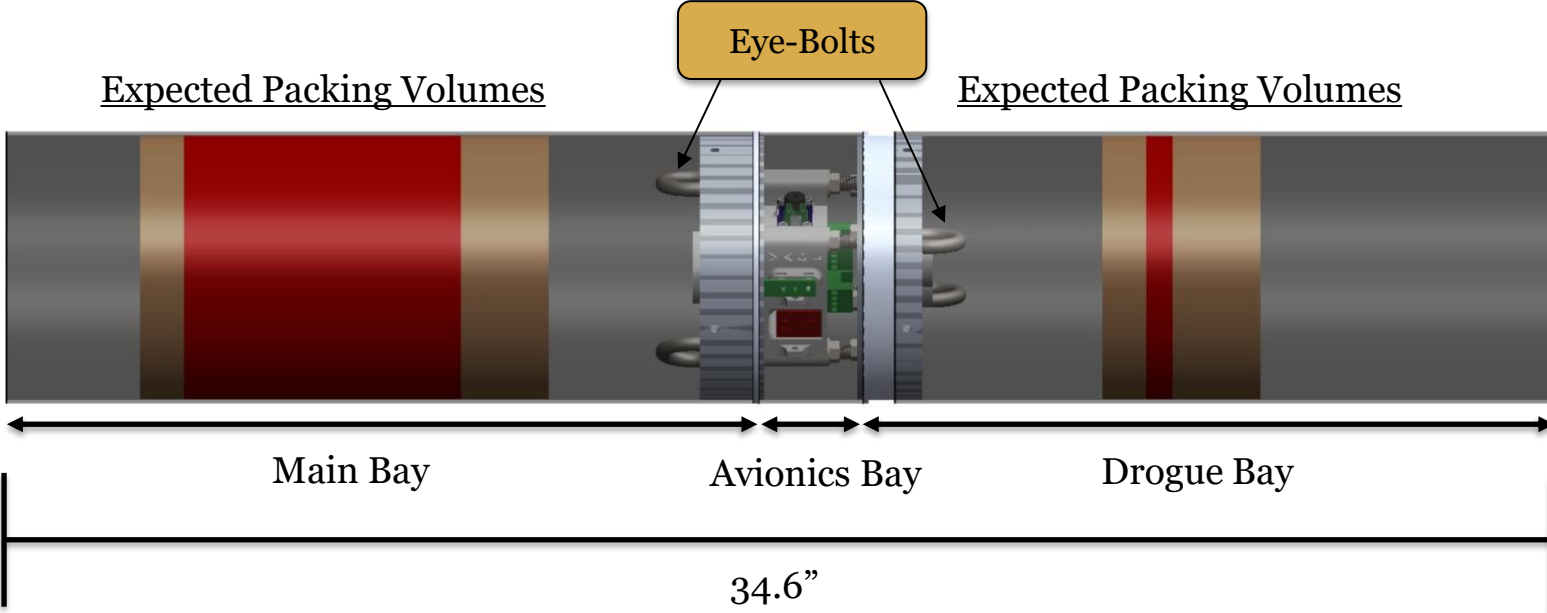
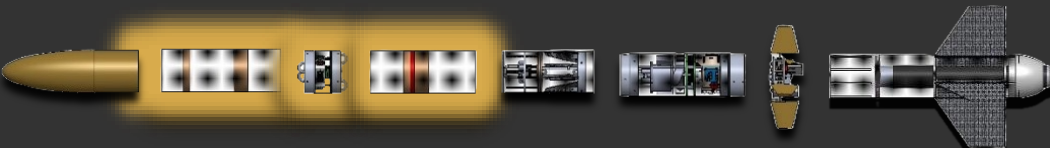


Figure 1: Recovery Section Components Overview

Section Mass	Section Length	Landing Kinetic Energy	Drogue Shock Cord Length	Main Shock Cord Length	Coupling
13.6 lbm	34.6"	33.5 ft-lbf	12 ft	16.5	4 x 2-56 nylon shear pins

Table 1: Recovery Section Statistics





Tail Section Overview

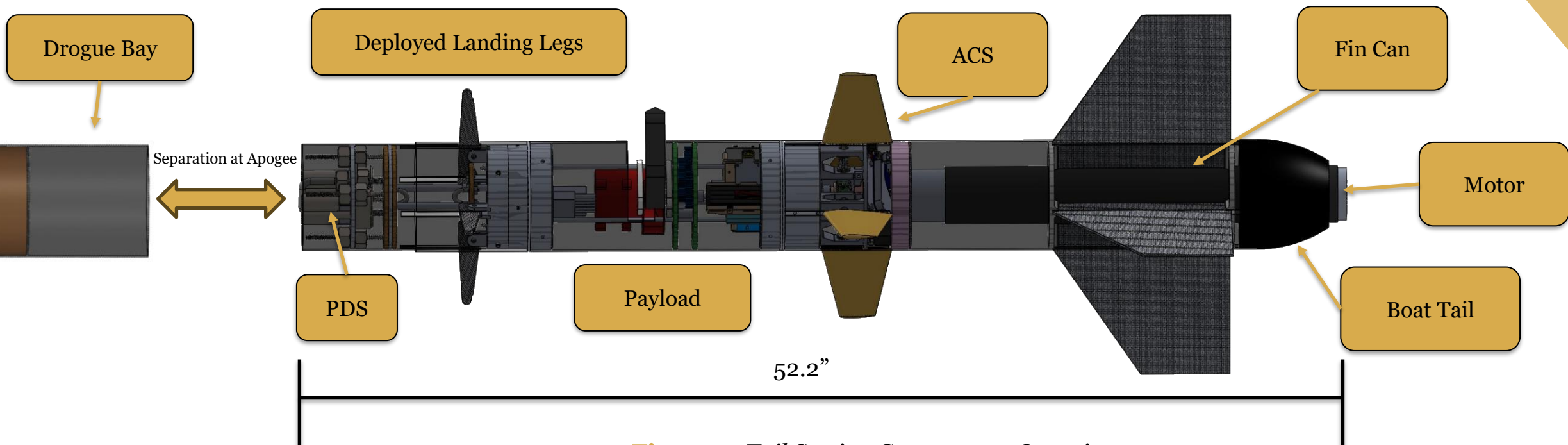


Figure 1: Tail Section Components Overview

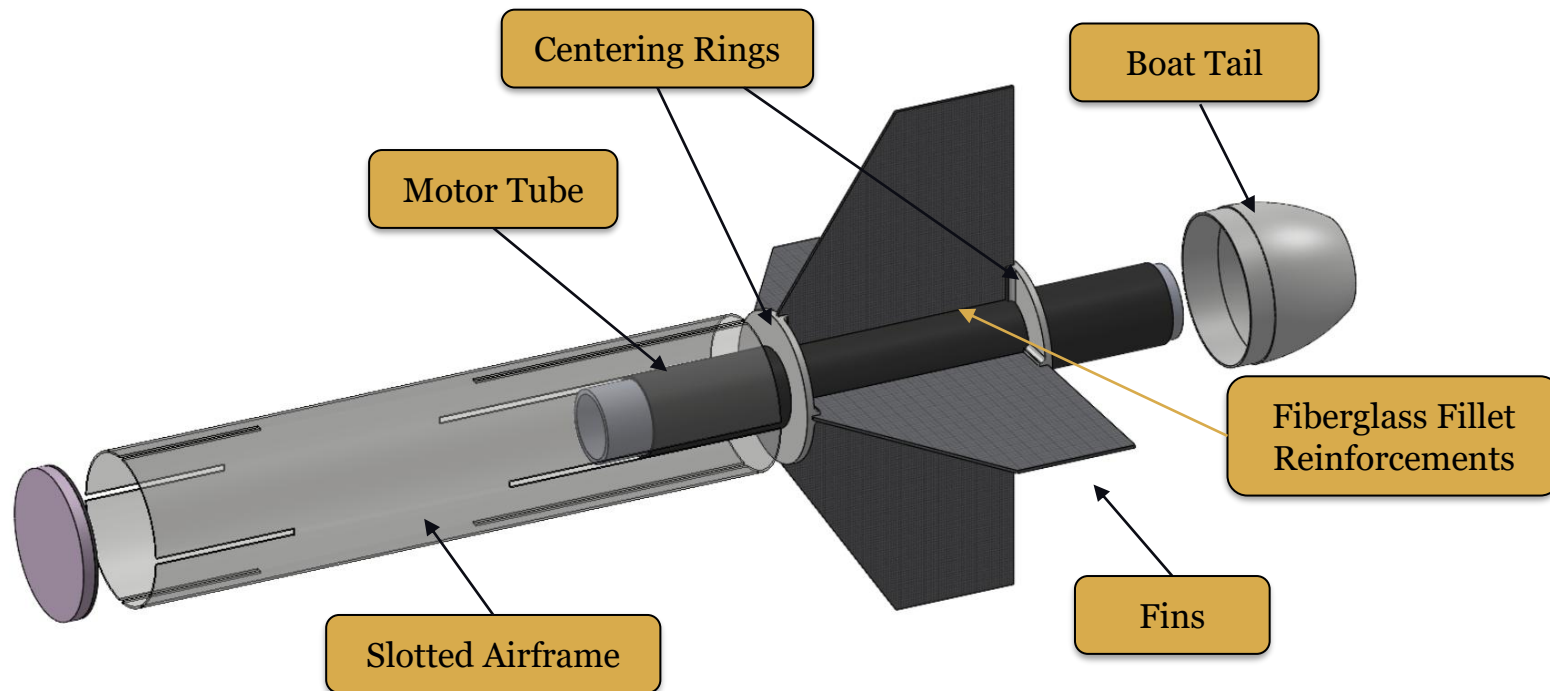
Section Mass	Section Length	Landing Kinetic Energy	Shock Cord Length	Coupling
28.1 lbm	52.5"	69.4 ft-lbf	26 ft	4 x 2-56 nylon shear pins

Table 1: Tail Section Statistics





Fin Can and Motor Retention



Manufacturing Procedure

1. Assemble the tail section alignment jig and line up all components.
2. Epoxy the fins to the motor tube
3. Add multi-layer fiberglass fillets on all eight fin-tube interfaces
4. Slide the Fin Can and Motor Retention Assembly into the slotted airframe. Add multi-layer fiberglass fillets

Figure 1: Exploded View of Fin Can and Motor Retention





Launch Vehicle Motor Selection

- Primary motor: **Aerotech L1940**
 - Requires small amount of ballast for 4200 ft ACS flight
 - High LRE velocity to improve aerodynamic and limit wind-cocking risk
- Backup motor: **Cesaroni L2375**
 - Utilized in case of excessive mass creep
 - Higher thrust needed to reach goal apogee

Parameter	Aerotech L1940	Cesaroni L2375
Total Impulse (lbf·s)	973	1093
Peak Thrust (lbf)	521	629
Average Thrust (lbf)	436	551
Burn Time (s)	2.2	1.9
Propellant Mass (lb)	4.02	5.12
Loaded Mass (lb)	8.50	9.17

Table 1: Primary and Backup Motor Overview

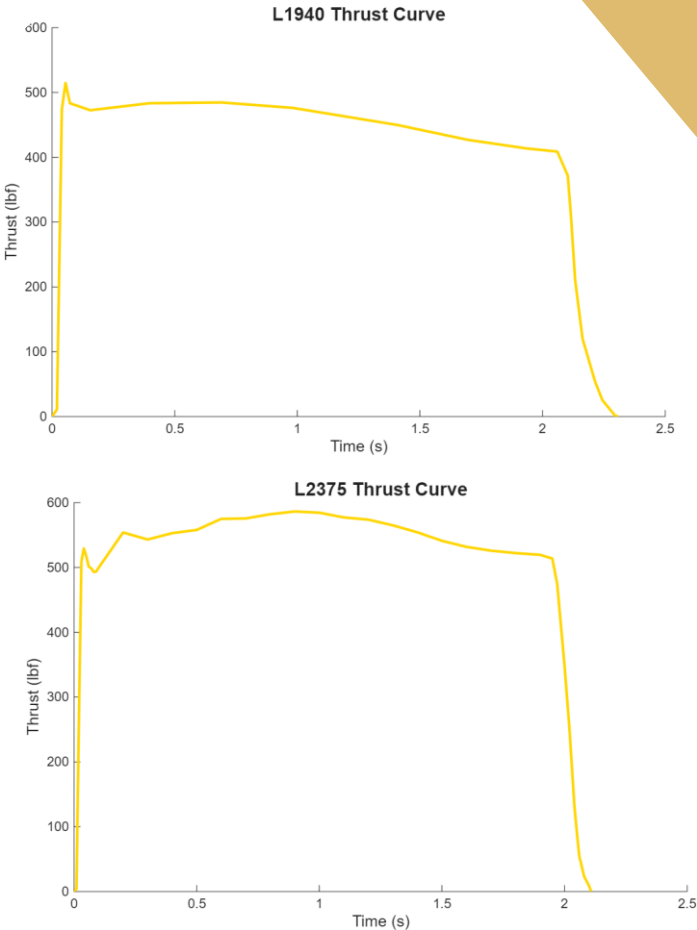


Figure 1: Primary (Top) and Secondary (Bottom) Motor Thrust Curves



Leading Recovery Design



Separation Points and Landing Sections

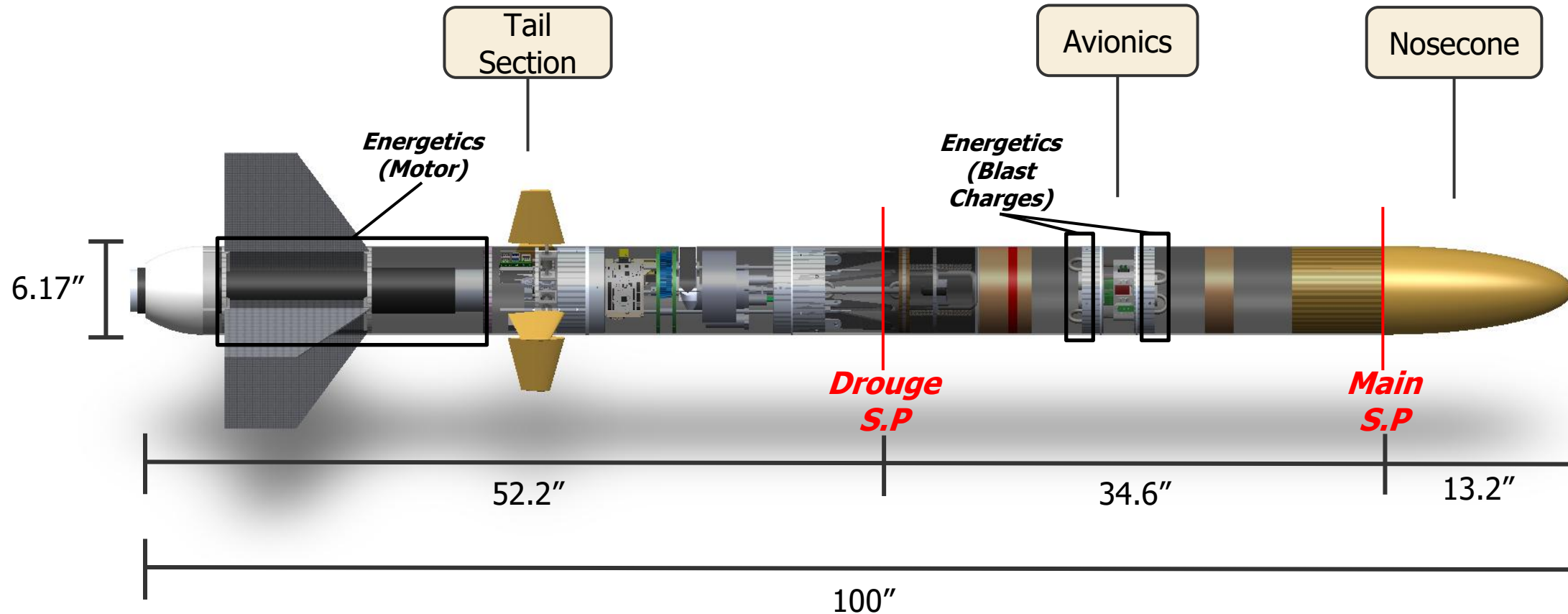


Figure 1: Separation Points Outlined on Rocket CAD

Dual Deployment Overview



Drogue: Fruity Chutes 24" Elliptical

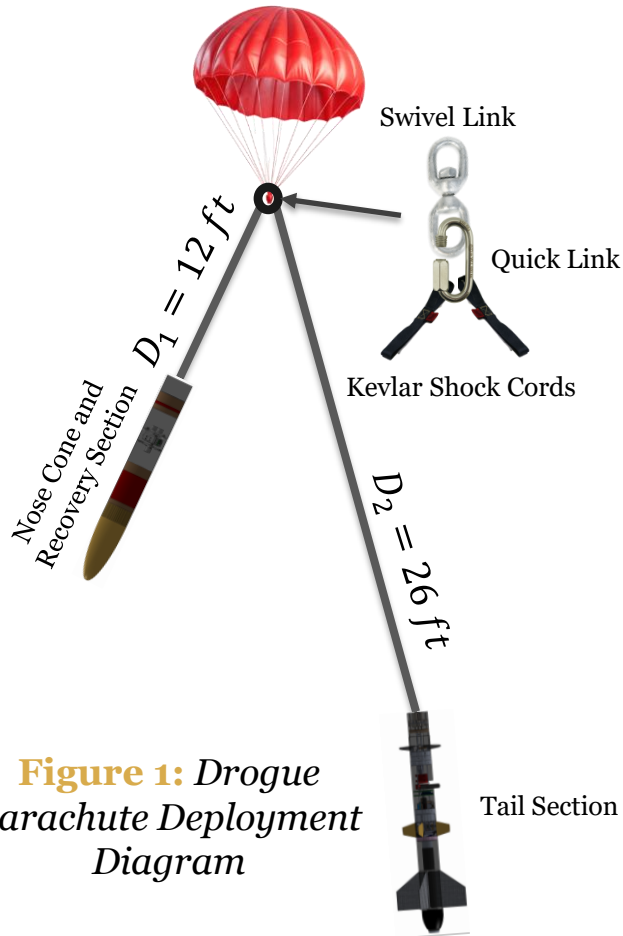


Figure 1: Drogue Parachute Deployment Diagram

Dual-deployment strategy ensures controlled descent and minimal drift.

Drogue parachute: Deployed at apogee to slow and stabilize vehicle.

Main parachute: Deployed at **550 ft AGL** for safe touchdown.

Main: Fruity Chutes Iris Ultra
144" Toroidal

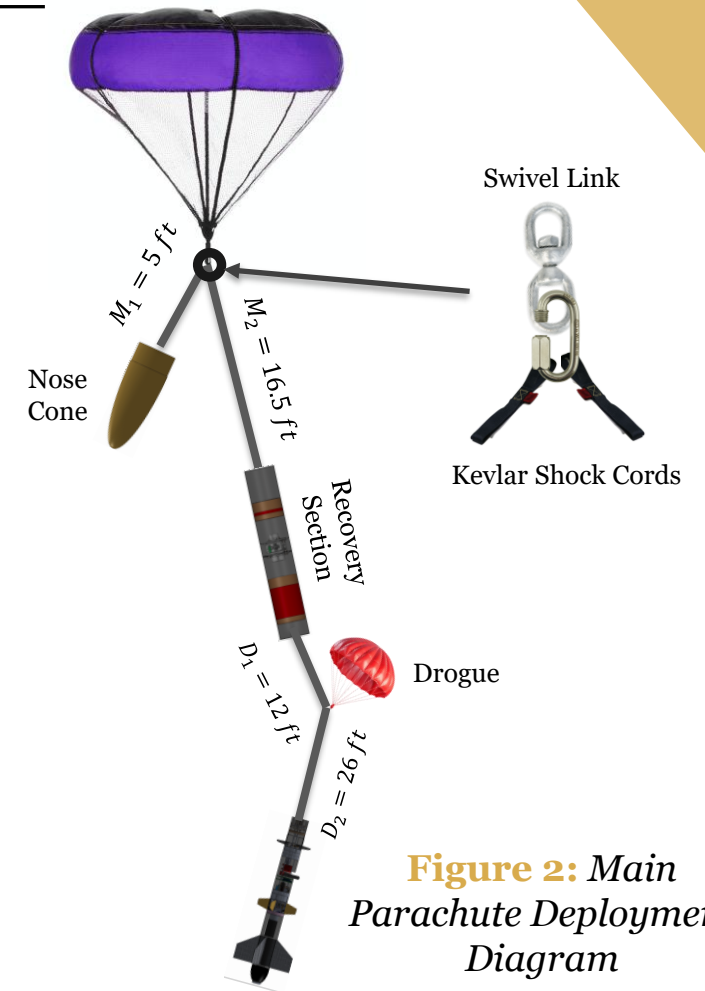


Figure 2: Main Parachute Deployment Diagram



Energetic Materials

Energetics Sizing Method:

- I. Ideal Gas Law
- II. Shear Relationship (Shear Pins and Pressure)

Underlying Equations

$$m = \frac{PV}{RT}$$

$$P = \frac{nF}{A}$$

Solving for a Safety Factor of 4...

Event	Safety Factor	Primary (g)	Backup (g)
Drogue	4	2.07	2.48
Main	4	4.53	5.44

Table 1: Energetics Mass for Parachute Deployment

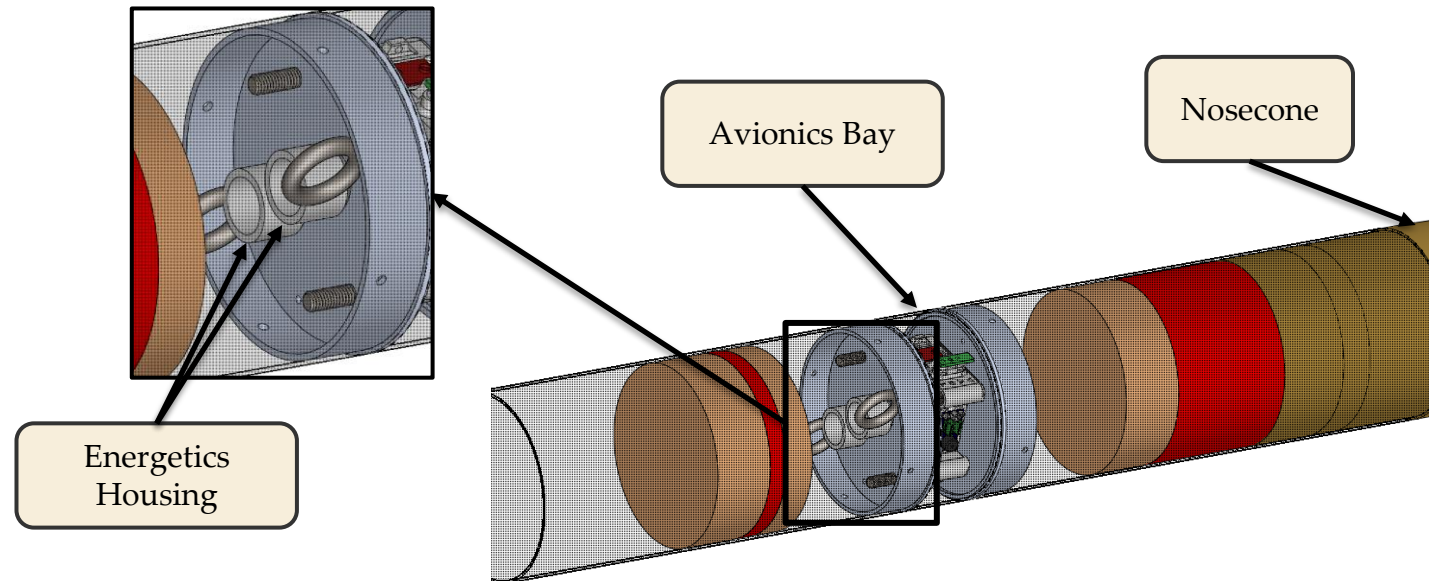


Figure 1: Energetics Housing Location



Altimeter Selection and Alternatives

Selected Altimeter

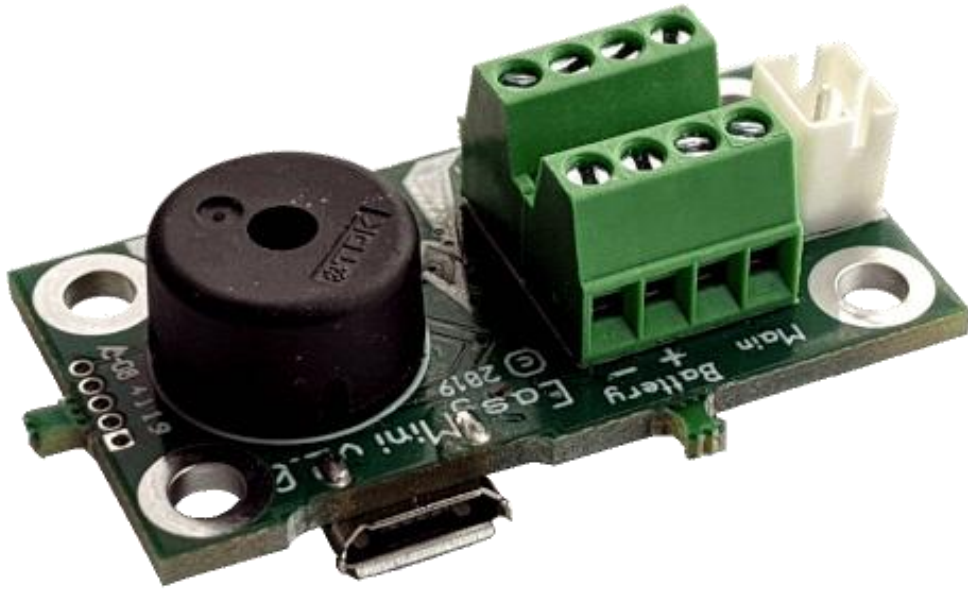


Figure 1: Altus Metrum EasyMini Altimeter

- Previously integrated
- Small and lightweight
- Commercially available
- 100 Hz sampling
- Runtime > 4 hours

Alternative Altimeters

- Not owned
- Low data logging frequency

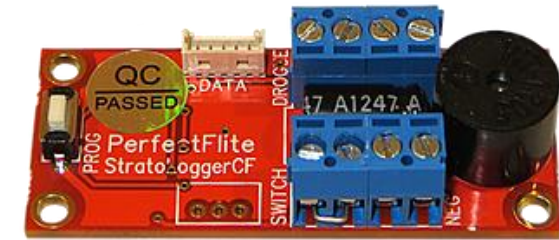


Figure 2: PerfectFlite Stratologger CF

- Mixed performance
- Profile is too big

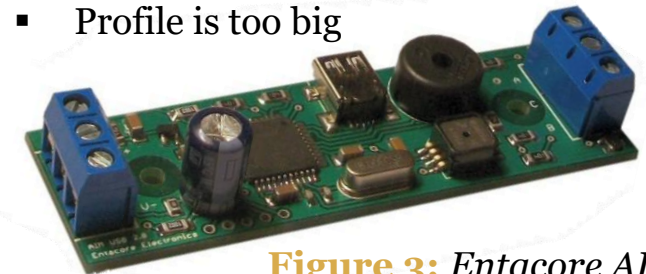
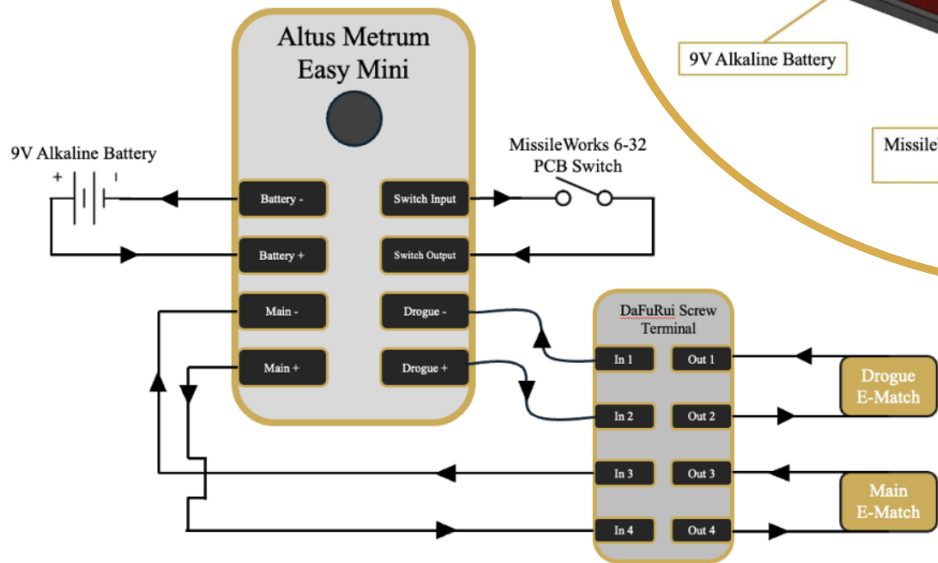


Figure 3: Entacore AIM USB

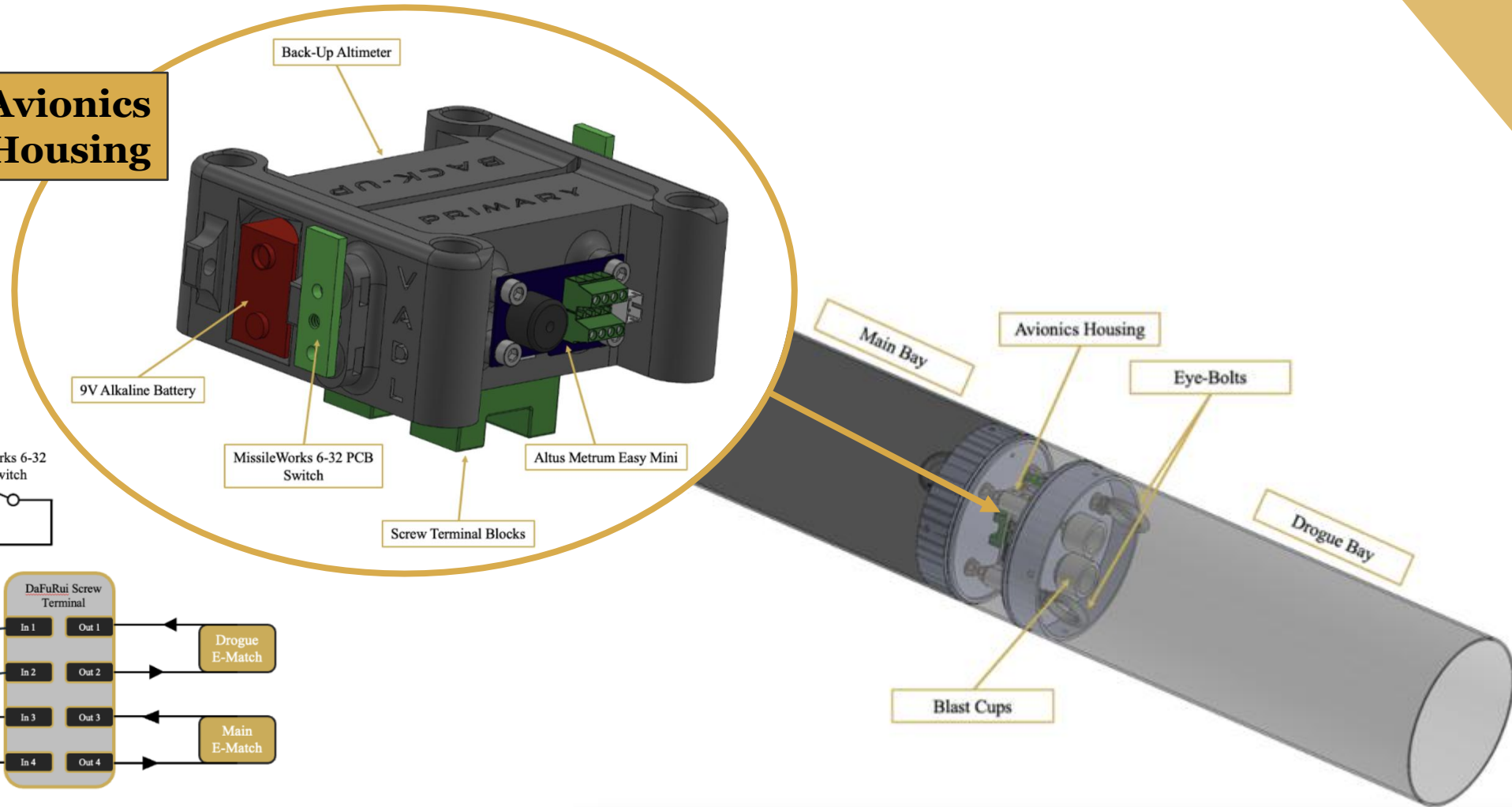


Avionics Housing and Electronics

Avionics Circuitry



Avionics Housing



GPS Tracker



Selected GPS Tracker

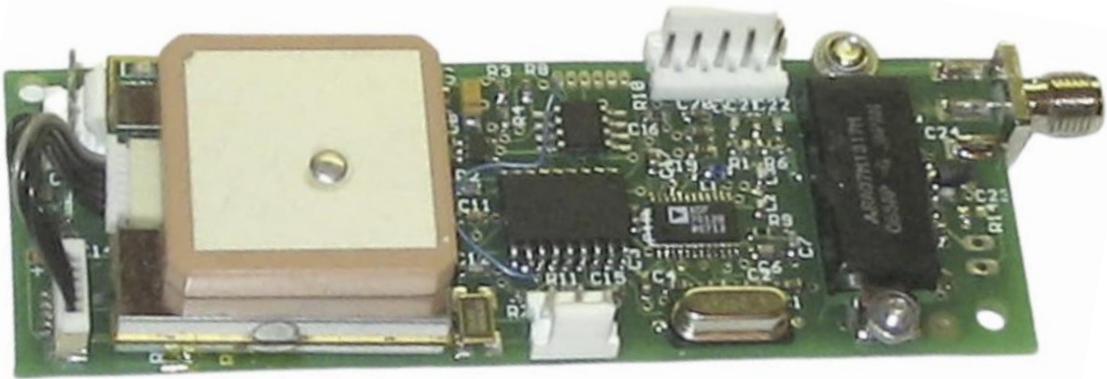


Figure 1: BigRedBee APRS Tracker

BigRedBee APRS Tracker	
Operating Frequency	420–450 MHz (70 cm amateur radio band)
RF Power Output	100 mW
Expected Runtime	≈ 10 hours
GPS Logging Rate	1 Hz (internal memory)

Table 1: BigRedBee APRS Statistics

Alternative GPS Trackers



Figure 2: Atlas Metrum Telemetry v2.0



Figure 3: Featherweight GPS Tracker



Figure 4: Missile Works WTX-T3

Drogue and Main Parachutes



Figure 1: 24" Drogue Fruity Chutes Elliptical Parachute



Figure 2: 144" Main Fruity Chutes Toroidal Parachute

Table 1:
Parachute
Specifications

Parachute	Type	Shock Cord	Cd	Packed Volume	Terminal Velocity	Anchor Hardware
Drogue	24" Fruity Chutes Classic Elliptical	38 (ft) of 1/2 in Kevlar	1.5	215 (in ³)	92.1 (in ³)	Eye-Bolts (Fore) U-Bolt (Aft)
Main	144" Iris Ultra Compact Toroidal	21.5 (ft) of 1/2 Kevlar	2.2	380 (in ³)	12.6 (in ³)	U-Bolt (Fore) Eye-Bolts (Aft)

Flight Predictions



Target Apogee and Flight Profile

Target Apogee: 4,200 ft

Predicted Apogees	No Wind No Cant	5 mph Winds 5° Cant
OpenRocket	4,349	4,275
MATLAB	4,401	4,317
RockSim	4,370	4,301

Table 1: Apogee reached on different simulation software's with ideal and expected conditions

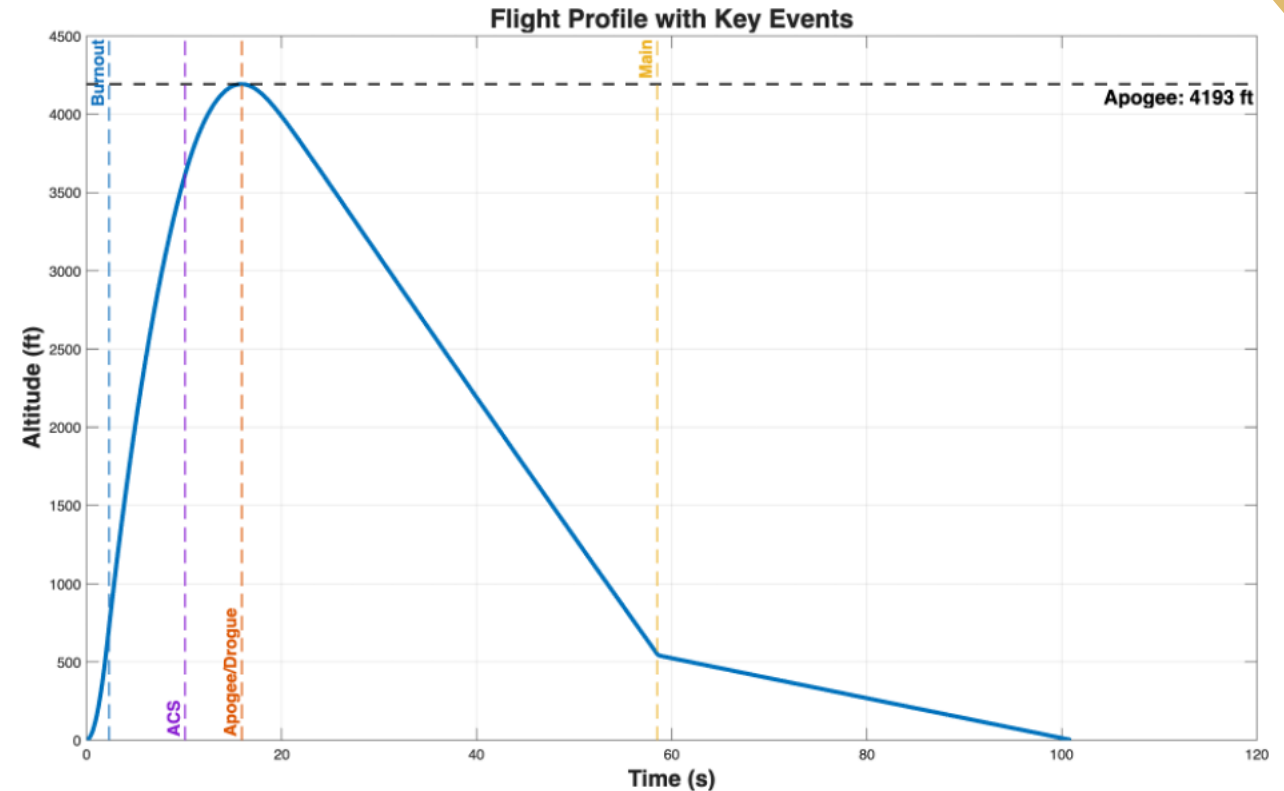


Figure 1: MATLAB simulation run on 5 mph winds, 5° cant and ACS deployment at ~3,650 ft AGL.
Apogee: 4,193 ft



Aerodynamics Analysis

Predicted Apogees	Pre ACS [0-3500 ft]	Post ACS [3500-4200 ft]
Avg. CP	72.4	72.4
Avg. CD	0.42	2.14

Table 1: Average Coefficient of Pressure (CP) and Coefficient of Drag (CP) before and after ACS deployment

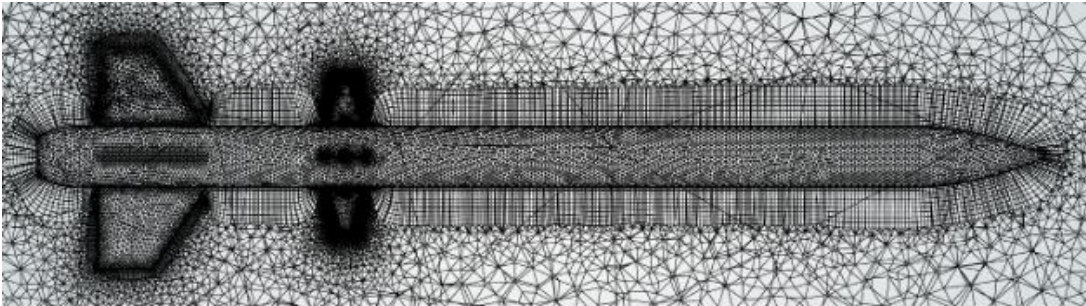


Figure 1: CFD Mesh of Fullscale Rocket

Figure 2: CD vs. Freestream Velocity, 0 deg ACS deflection. Average CD is 0.42

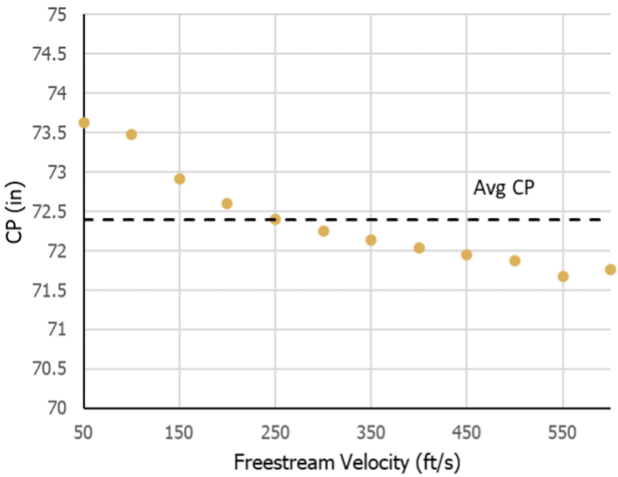
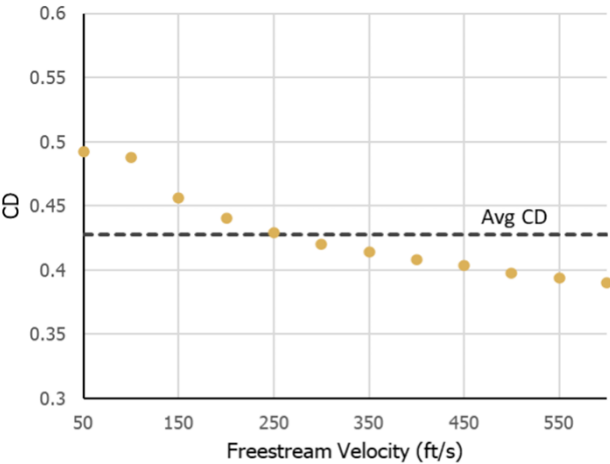


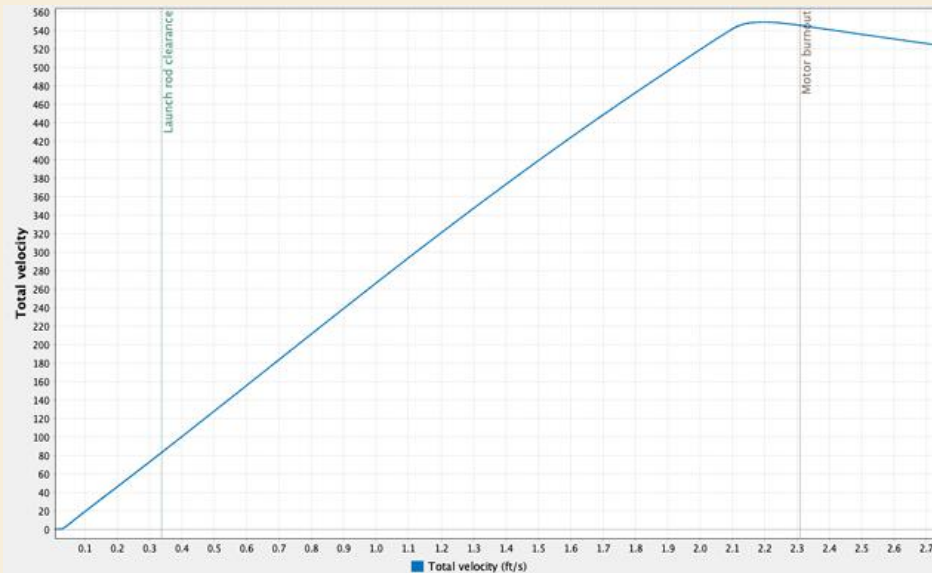
Figure 3: CP vs. Freestream Velocity, 0 deg ACS deflection. Average CP is 72.4 in.

Metric	Value
Avg CP	72.4 in
Avg CD (pre-ACS)	0.42



Thrust to Weight Ratio and Launch Rail Exit Velocity

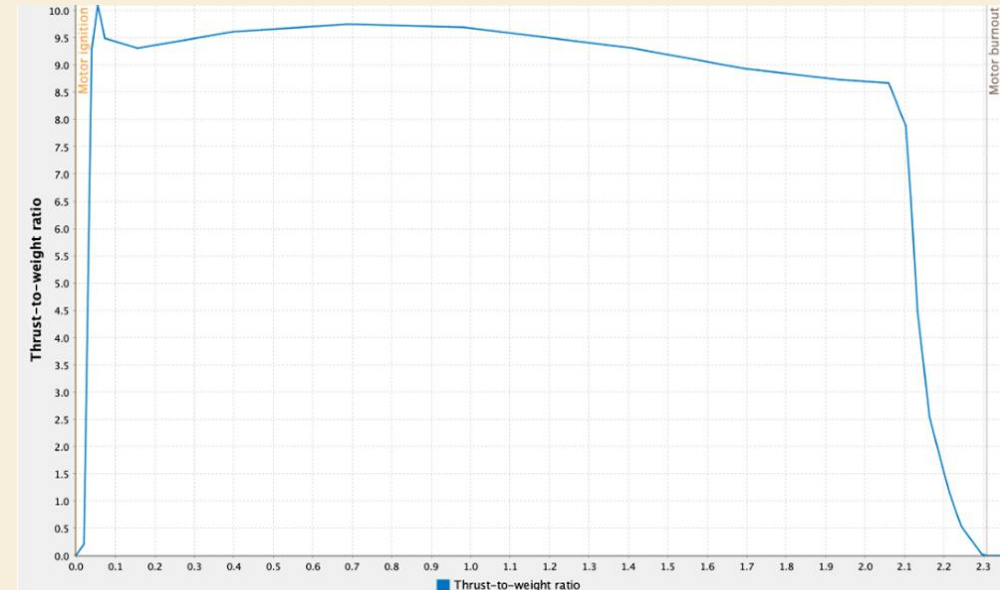
Motor Burn Velocity Profile



Motor	LRE Velocity (ft/s)	Max Velocity (ft/s)
Aerotech L1940 (Leading)	83.1	549
Cesaroni L2375 (Backup)	86.4	636

Table 1: Leading and Backup Motor Velocities

Thrust-to-Weight Curve



Motor	Average Thrust (lbf)	Vehicle Wet Mass (lbm)	Thrust-to-Weight Ratio
Aerotech L1940 (Leading)	83.1	549	8.53
Cesaroni L2375 (Backup)	86.4	636	10.64

Table 2: Leading and Backup Motor Thrust



ACS, Ballast, and Wind Affects on Apogee

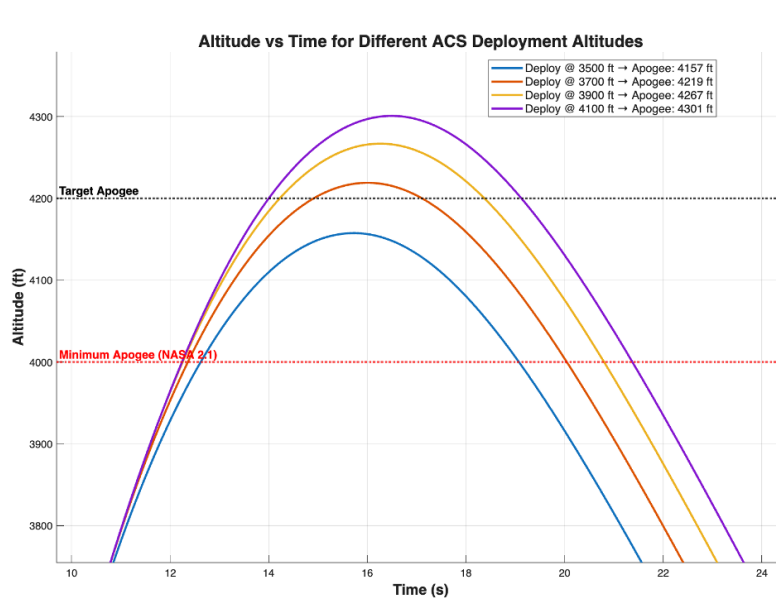


Figure 1: Apogee sensitivity to deployment

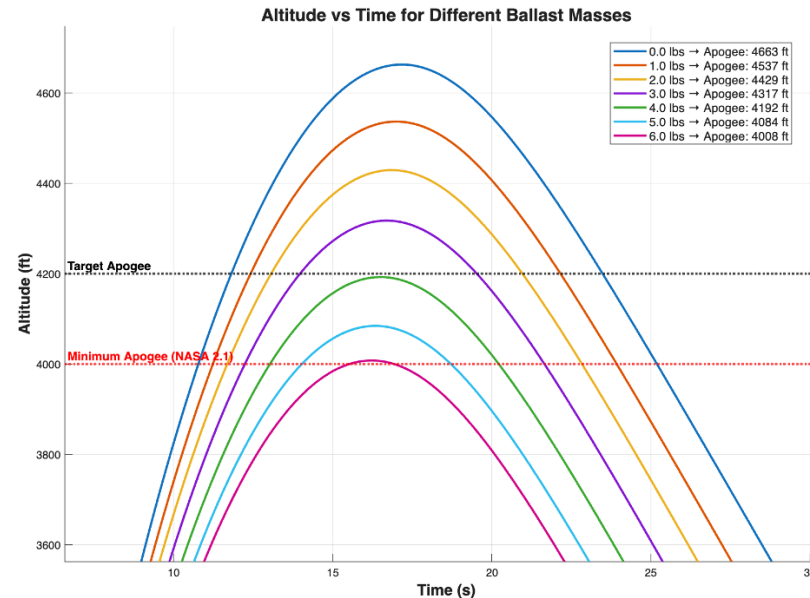


Figure 2: Apogee sensitivity to ballast mass

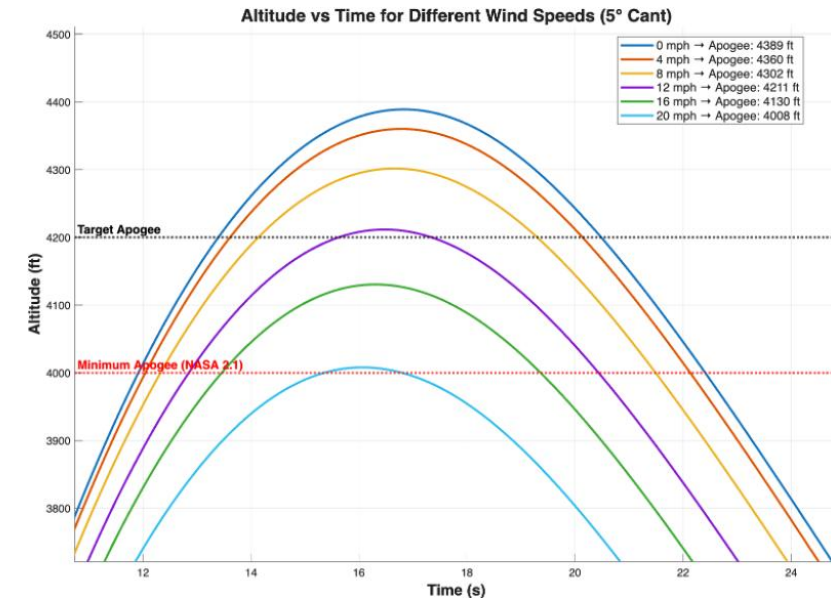


Figure 3: Apogee sensitivity to wind speed



Descent Time and Drift

- $t_{descent} = t_{drogue} + t_{main}$
 - $t_{drogue} = 41.6$ seconds
 - $t_{main} = 43.6$ seconds
 - $t_{descent} = 85.2$ seconds
- Worst case drift is 2,488 ft
 - With constant 20 mph winds
 - Still within NASA recovery radius

Wind (mph)	Drift (ft)
20	2,015
15	1,711
10	1,487
5	1,215
0	981

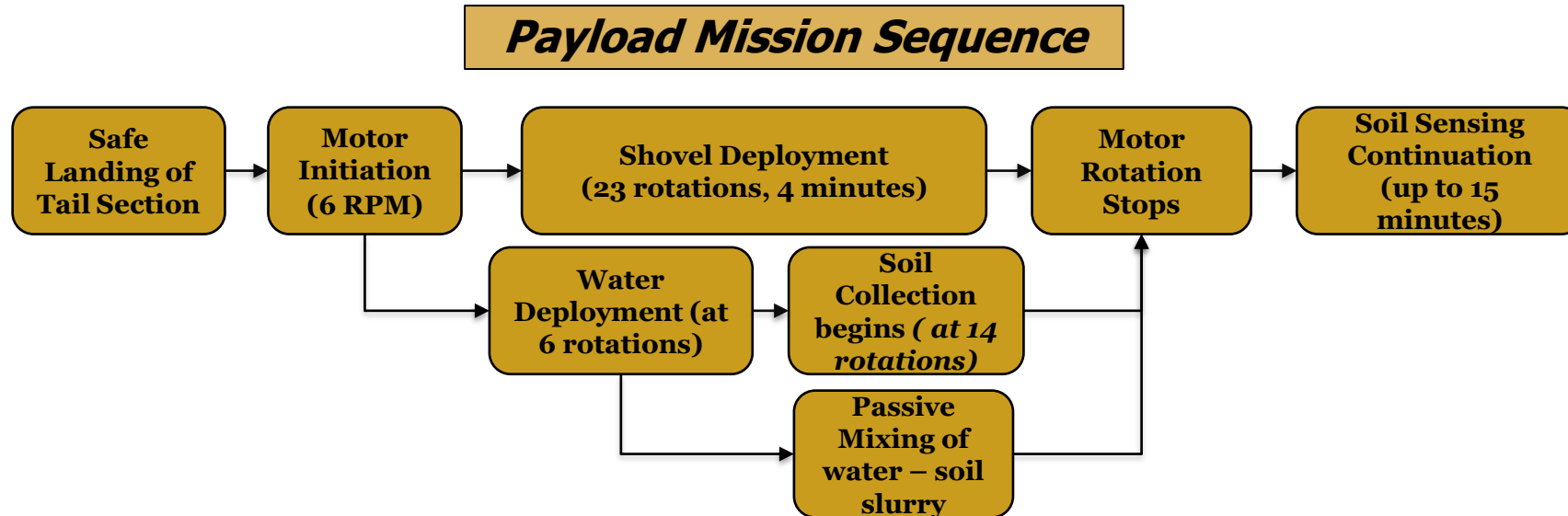
Table 1: Recovery drift at different wind speeds simulated on OpenRocket

Payload Design



Payload Mission Overview

After landing, the payload autonomously collects **100 mL of soil** and performs **in-situ analysis** of its **electrical conductivity, pH, and nitrate content**. A single motor drives a rotating shovel mechanism that digs and transfers soil into the sampling chamber. A passive water-release system then moisturizes the sample to enable accurate sensor measurements. All data is logged and time-stamped onboard to meet NASA verification requirements.





Payload Success Criteria

Subsystem	Shovel Opening	Water Injection	Electronics	General
Criteria	Soil collection begins autonomously with 23 payload rotations to deploy shovel and collect 100 mL soil	Water is released at a ratio of 1:1 water to soil or greater.	Sensors record conductivity, pH, and nitrate content accurately	All payload operations completed in <15 minutes
NASA Req.	NASA Req. 4.1	N/A	NASA Req. 4.2, 4.2.1	NASA Req. 4.1.1
VADL Req.	VADL Req. 3.2.1, 3.2.3	VADL Req. 3.4.3	VADL Req. 3.5.14	VADL Req. 3.2.2

Table 1: Abbreviated Success Criteria



Payload Bay Overview

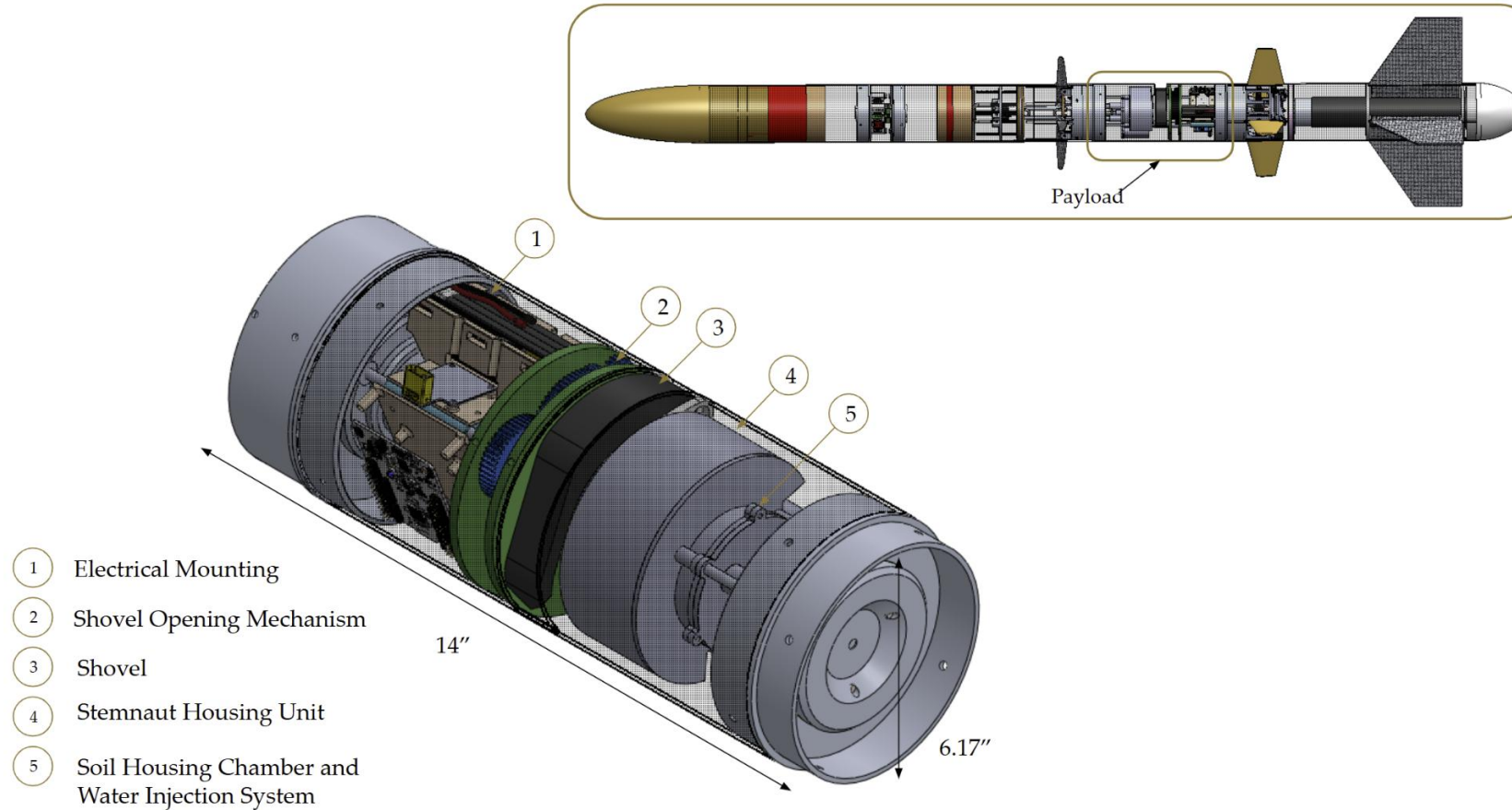


Figure 1: *Payload Bay Components*



Alternative Soil Extraction Designs

Rotating Shovel

Payload bay rotates causing shovel to extend and dig

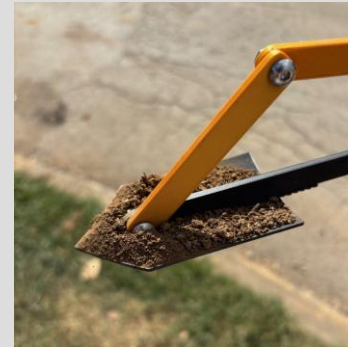
8.85/10



Horizontal Scraper

Linkage actuates digging in elliptical path

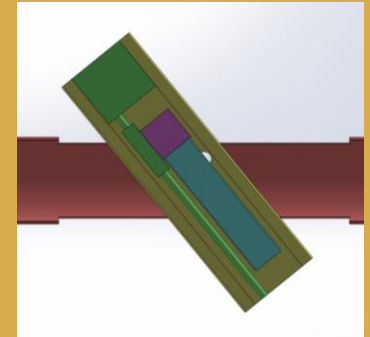
5.7/10



Non-Jettisoned Auger

Payload reorients to drill soil from horizontal landing

5.3/10



Step Bit Auger

Extends with gear pushing and spinning threaded rod

5.7/10



Rotating Buckets

Buckets move along conveyer scooping soil

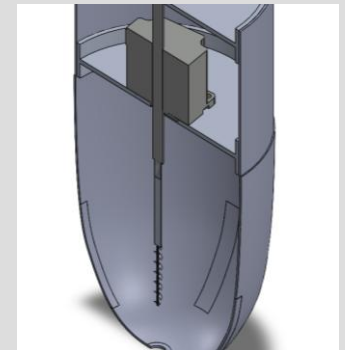
5.3/10



Jettisoned Auger

Nosecone auger actuated by motor turning miter gears

4.2/10





Shovel Extension Mechanism

We decided to use a sandwiched gearbox mechanism for shovel extension due to its efficacy and robustness.

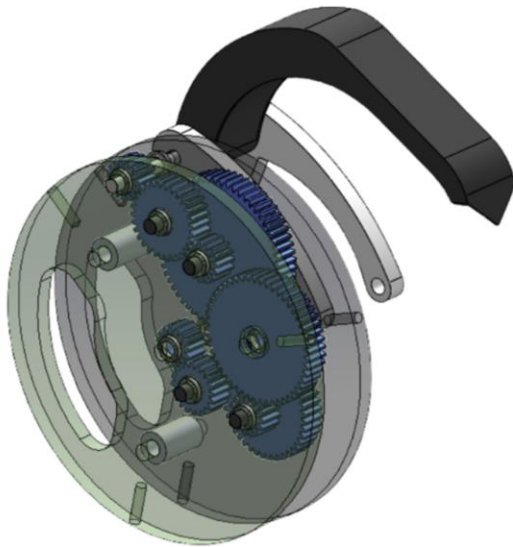


Figure 1: *Shovel Extension Mechanism Deployed*

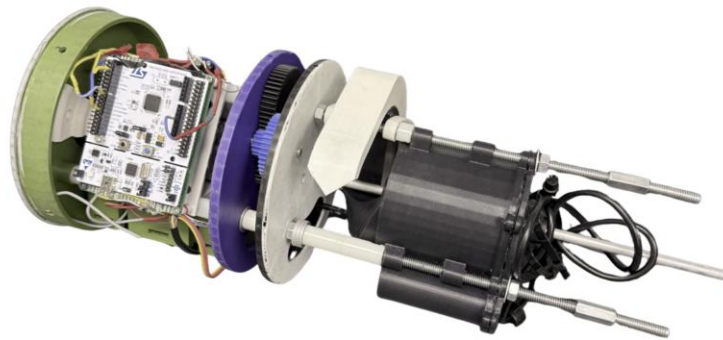


Figure 2: *Prototype Payload with Integrated Shovel Extension Mechanism (White and Blue)*

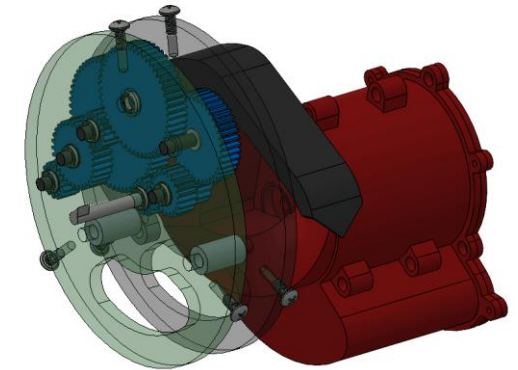


Figure 3: *Shovel Extension Mechanism Deploying*

Soil Housing Design

Soil Housing Specifications	
Parameter	Value / Description
Material	ABS (vapor-smoothed)
Usable Internal Volume	≈ 250 mL
Soil Capacity	100 mL (meets NASA Req. 4.1, & VADL Req. 3.2.1)
Maximum Water Capacity	150 mL (for 1:1.5 soil-to-water ratio)
Factor of Safety (Volume)	2.0

Table 1: Soil Housing Specifications

Soil Entry Mechanism Trade Study					
Design	Performance	Reliability	Soil Retention	Soil Friction	Total Weighted Score
Soil Slide	5	6	5	7	5.65
Spring-loaded Door	4	4	7	5	5.1
Actuated Door	6	4	7	3	5.2

Table 2: Soil Entry Trade Study

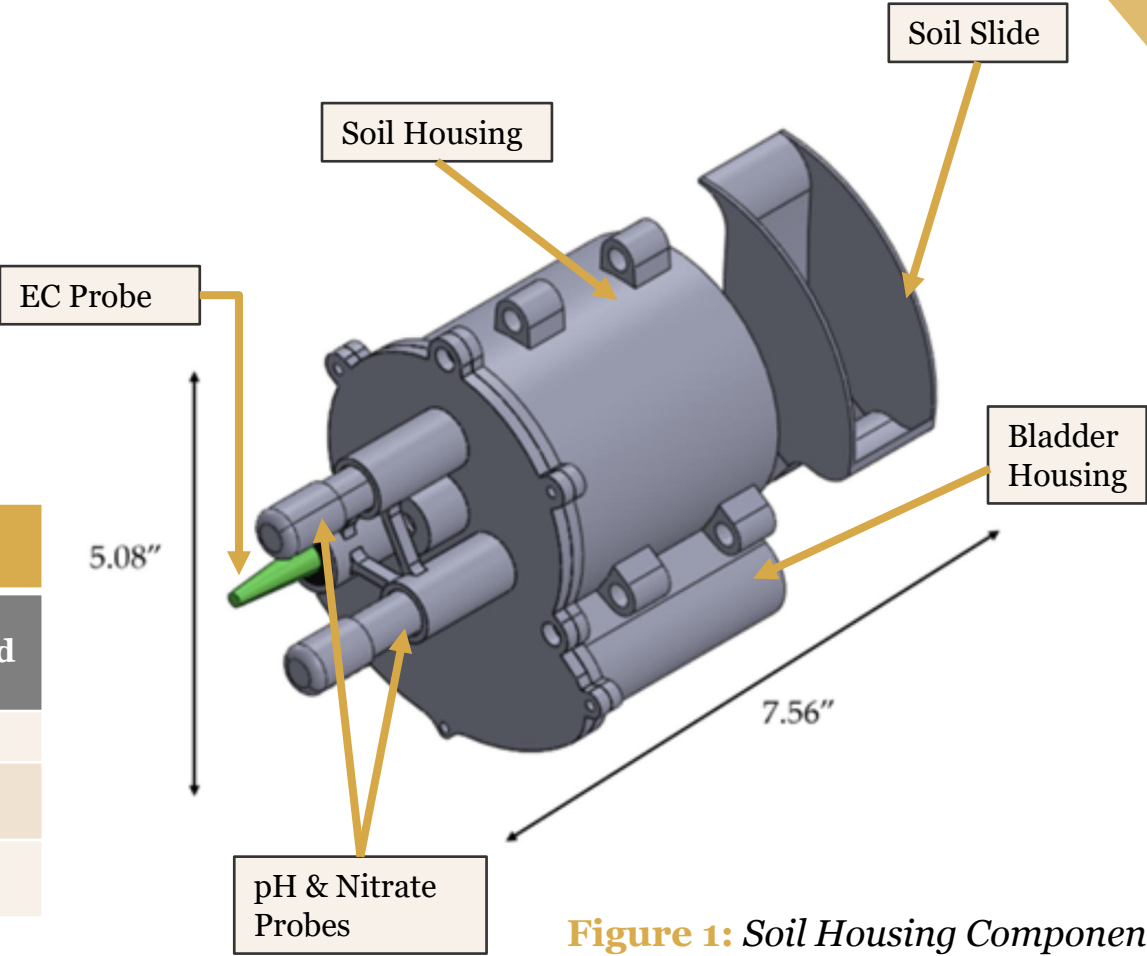


Figure 1: Soil Housing Components



Water Injection System Alternatives and Design

Water Integration Trade Study					
Design	Reliability & Leak Risk	Mass Impact	Mechanical Complexity	Release & Timing Control	Total Weighted Score
Syringe Injection	5	3	4	9	5.7
Tether-Released Bladder	6	7	8	5	6.25
Integration Chamber-Water System	4	8	6	3	4.7

Table 1: Water Integration Trade Study

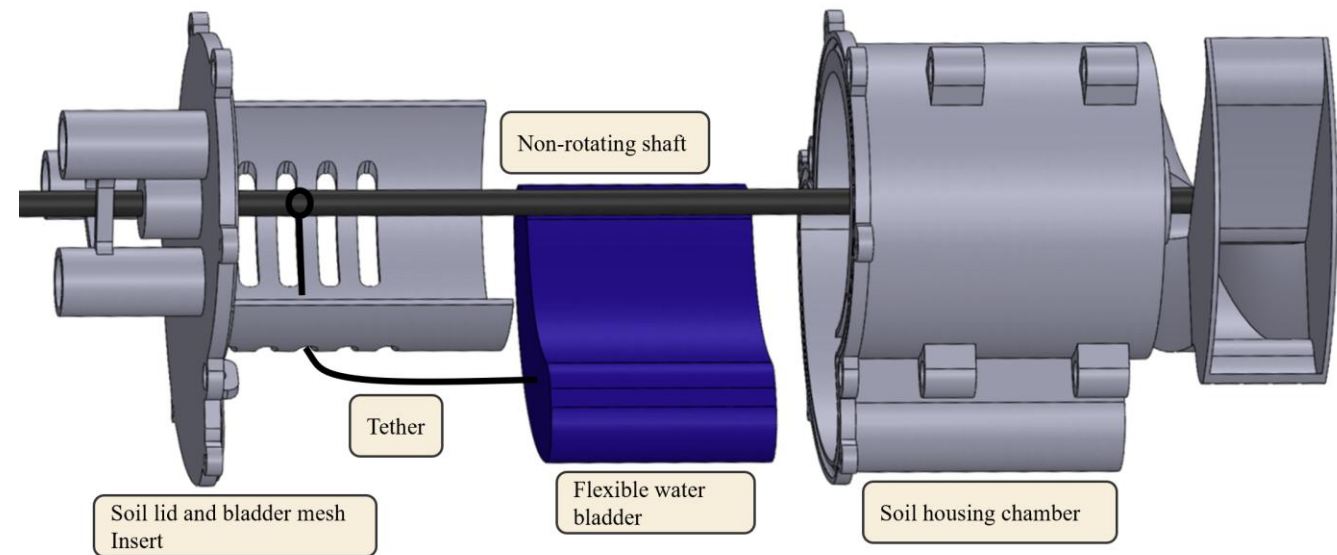


Figure 1: Water Exploded View of Tether-Released Bladder System



STEMNaut Housing

The **STEMNaut Housing** is a **centrifugal habitat** located around our soil collection chamber to allow our STEMNaut's **access to soil research**.

Parameter	Description
Internal Volume	40 in ³
Habitat Scale Ratio	1:40–1:50 (relative to full-scale habitat)
STEMNaut Height	≈ 0.5 in

Table 1: *STEMNaut Housing Statistics*

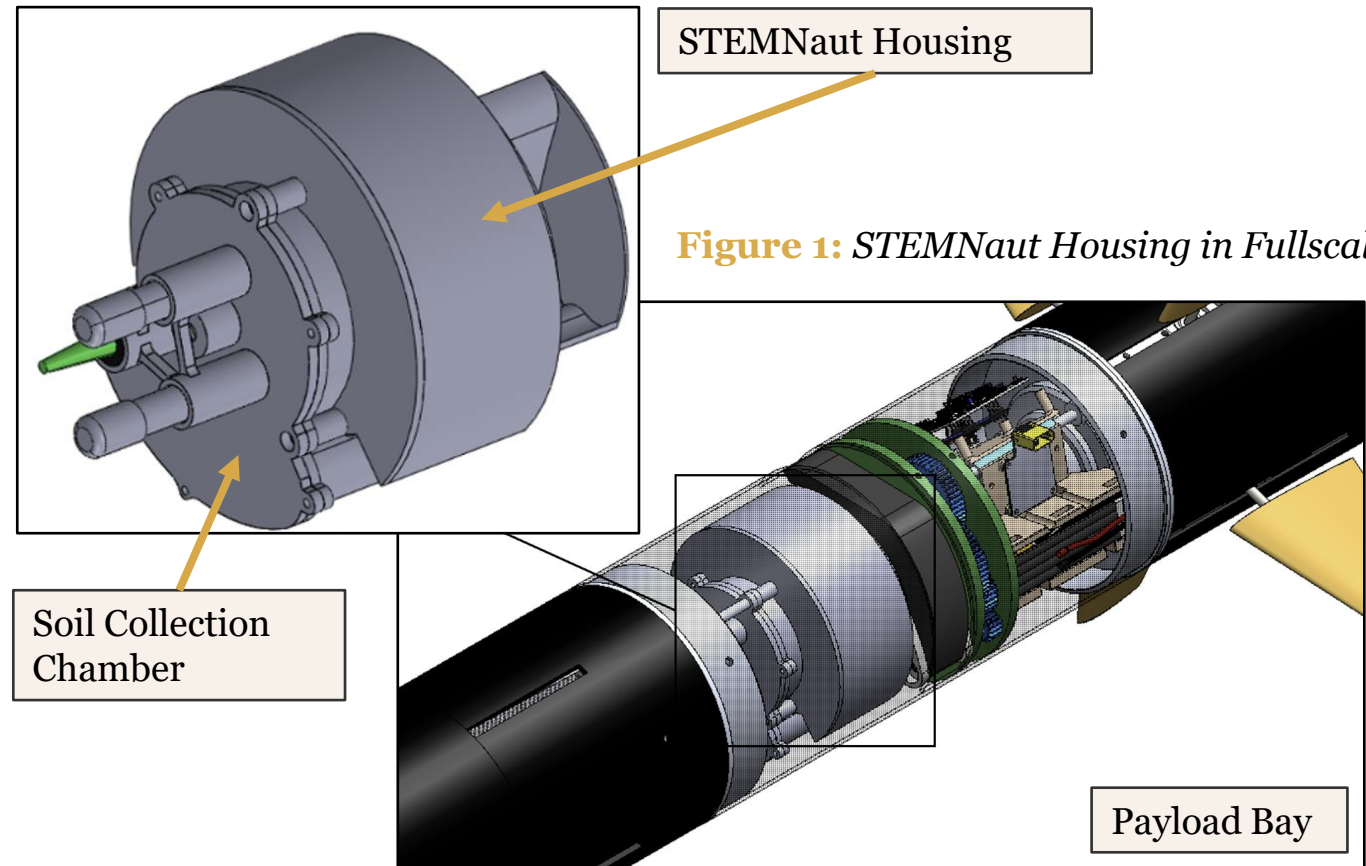


Figure 1: *STEMNaut Housing in Fullscale Design*



Sensor Suite Selections



Figure 1: *Conductivity Probe*

Atlas Scientific Mini Conductivity Probe

- Durable and Compact Design
- 5 – 200,000 $\mu\text{S}/\text{cm}$
- $\pm 2\%$ accuracy



Figure 2: *pH Probe*

NT Sensors pH Ion Selective Electrode

- PVC Membrane for In-flight Durability
- 0 – 14 pH
- $\pm 4\%$ accuracy



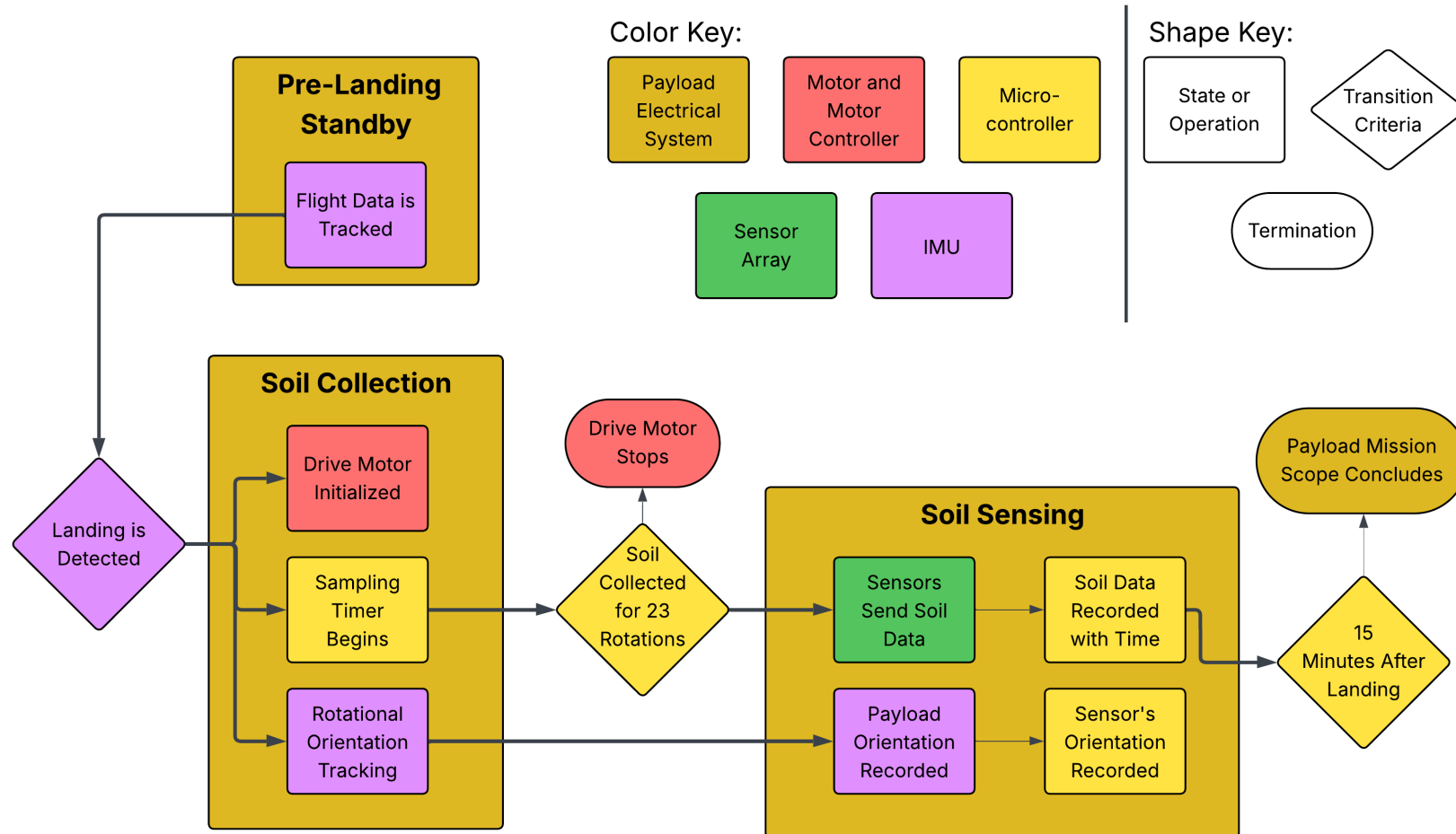
Figure 3: *Nitrate Probe*

NT Sensors Nitrate Ion Selective Electrode

- PVC Membrane for In-flight Durability
- 1 – 62 mg/L
- $\pm 4\%$ accuracy



Payload Electronics Logic





Payload Electronics Schematic

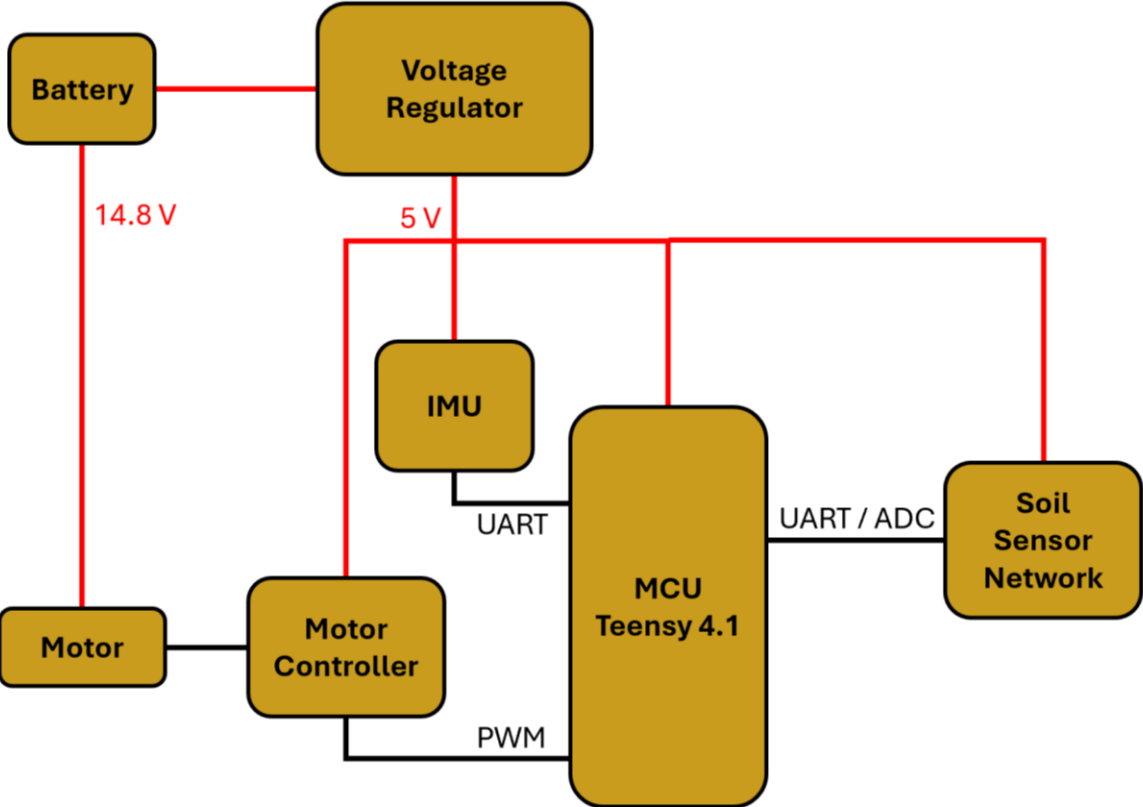


Figure 1: Payload Power and Control Diagram

Component	Selection
Microcontroller	Teensy 4.1
IMU	Vector NAV VN-100
Motor Driver	HW-039
Motor	42GP-775 DC Motor
Voltage Regulator	Pololu D36V50F5
Battery	14.8V LiPo Battery
Soil Sensors	NTSensor Electrode Atlas Scientific EZO™

Table 1: Payload Electronics Specifications



Payload Verification Plan

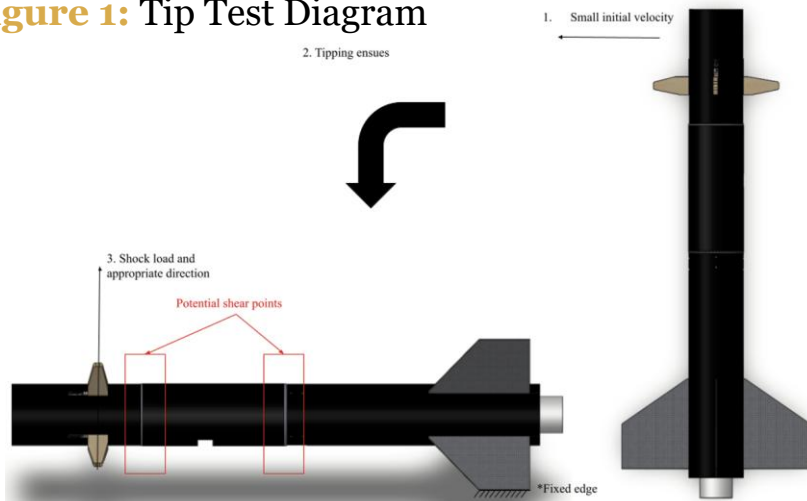
Tip Test
Verify autonomous actuation and soil collection of gearbox opening mechanism. Also, text that support legs/fins disallow payload to flip

Figure 2: Drop Test Rig



Payload Demonstration Flight
Ensure payload withstands full flight loads like parachute shock and that water integration and soil collection work repeatedly. IMU signal triggering opening mechanism and soil test data logging will also be verified

Figure 1: Tip Test Diagram



Drop Test
Verify operation of gearbox mechanism and water integration with landing KE stressing weaker parts

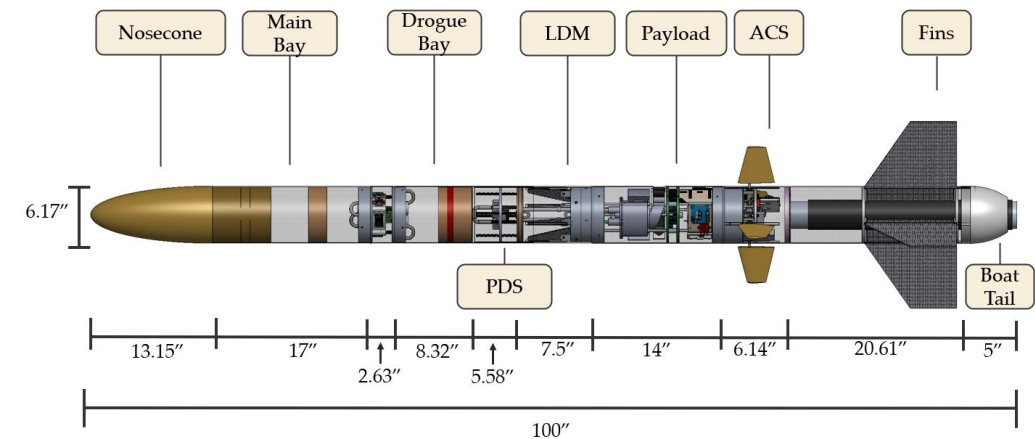


Figure 3: Full-Scale CAD

Apogee Control System (ACS)

ACS Success Criteria



Criterion #	Description	Relevant Requirements
1	Detect stages of flight using redundant acceleration and altitude data. ACS deployment occurs after MECO and 3,500 ft AGL.	VADL 2.4.3
2	Increase the drag coefficient (C_D) to at least 2 in reference to the cross-sectional area of the rocket.	VADL 3.6.1
3	Achieves target apogee within 5% error.	VADL 3.6.2
4	Maintains a static stability margin greater than 2 when fins are deployed.	VADL 2.4.1
5	ACS has minimum effect on ascent prior to deployment.	VADL 2.4.1 VADL 2.4.2 VADL 2.4.6

Table 1: PH Numerated ACS Success Criteria

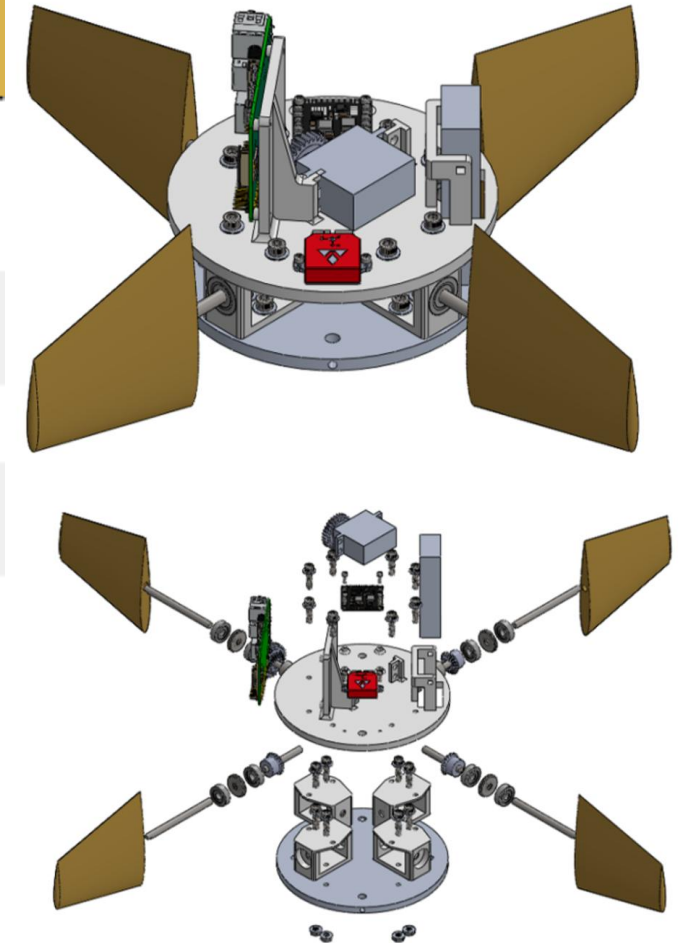


Figure 1: Assembled and Exploded ACS View



ACS Fin Design – Geometry and Materials

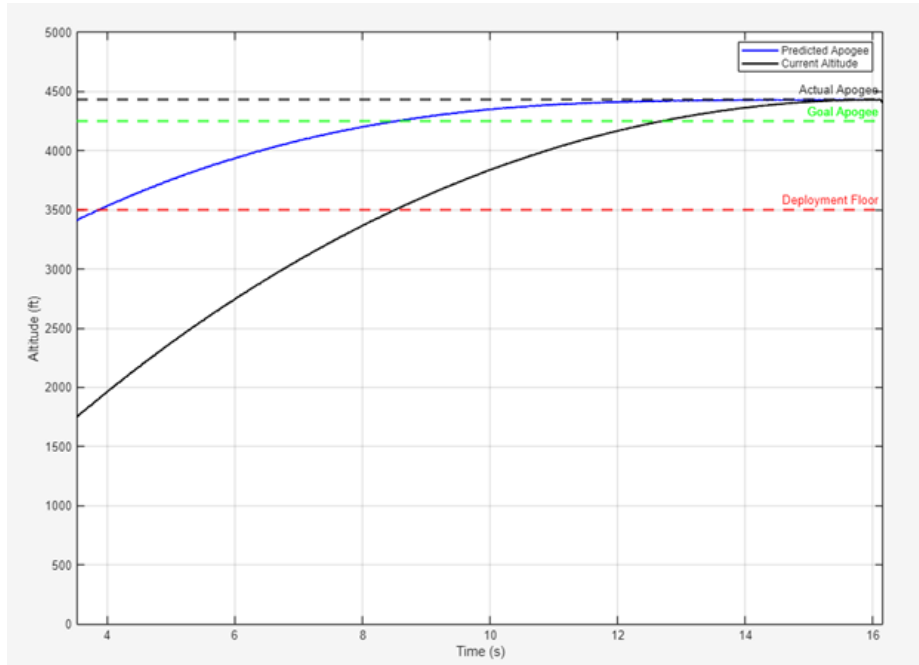


Figure 1: Apogee Prediction Graph

- To influence rocket over 200 ft deploying after 3500 ft, CD of 2 required
 - CD in reference to nosecone frontal area
- Area per deployed fin: 12 in²
- NACA 0012 Airfoils (Profile right)

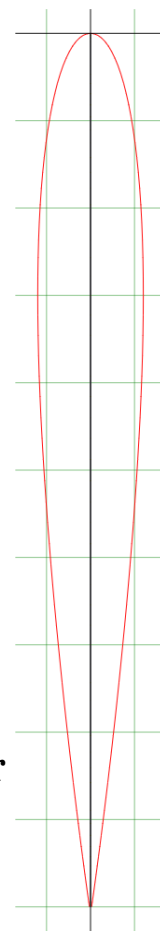


Figure 2: ACS Fin Profile

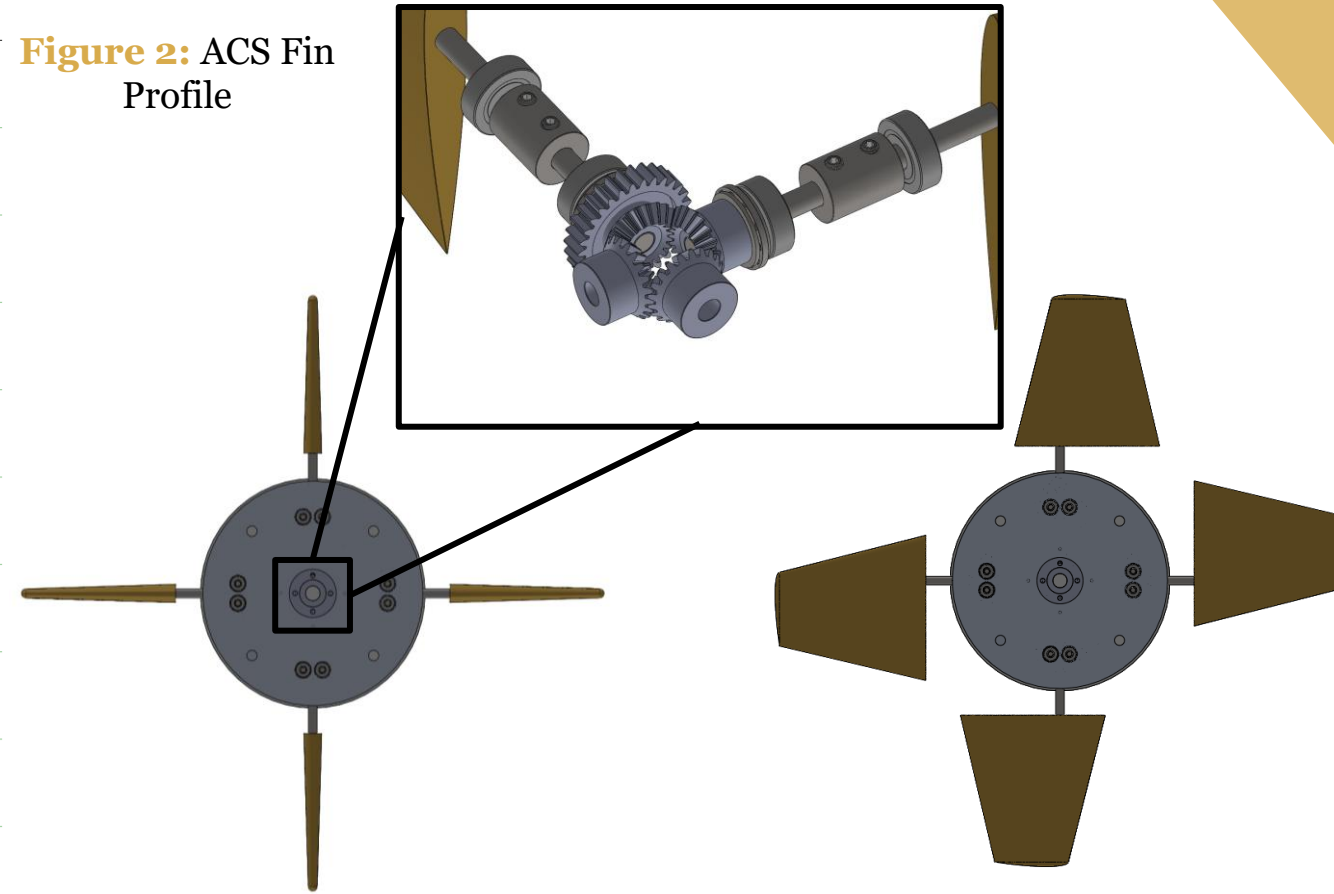


Figure 3: ACS Gearing Mechanism and Deployment

ACS Electronics and Flight Logic

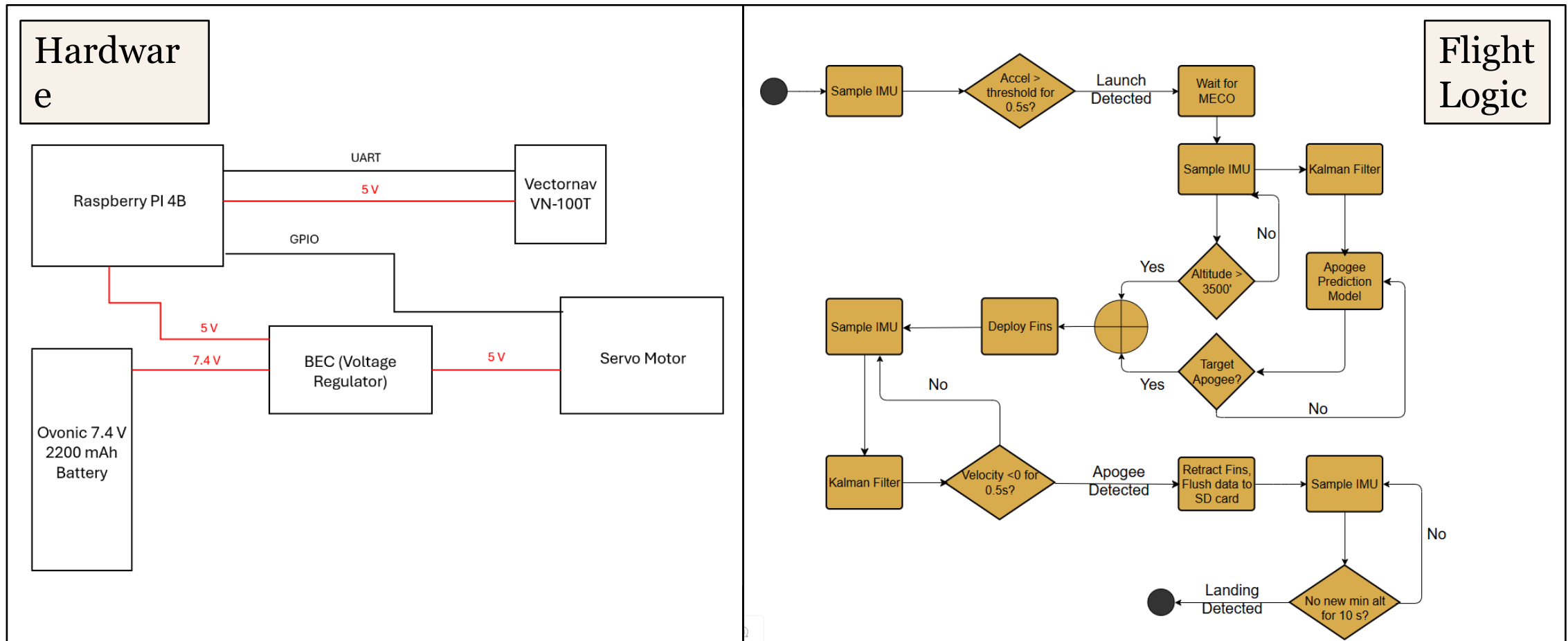


Figure 1: ACS Power and Communication Schematic

Figure 2: ACS Flight Logic Diagram



ACS Verification Plan

Verification Method

Ansys CFD Analysis

Wind tunnel Characterization

Hardware in the Loop (HIL)
Testing

- JSBSim software simulation
- Microcontroller interface

Table 1: ACS Verification Methods

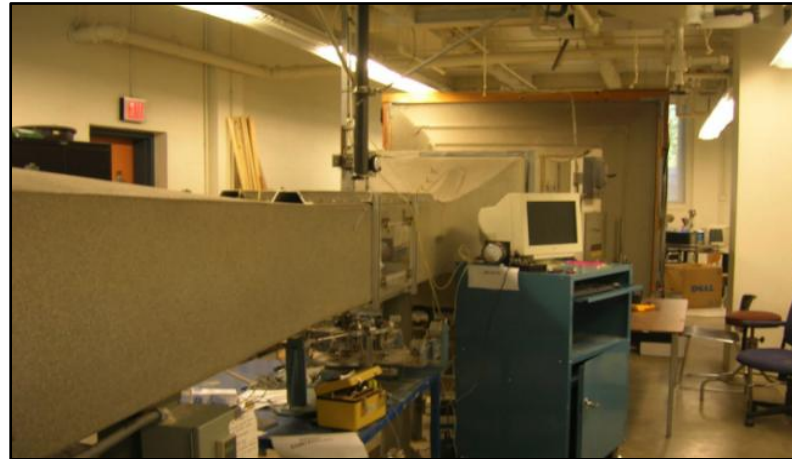
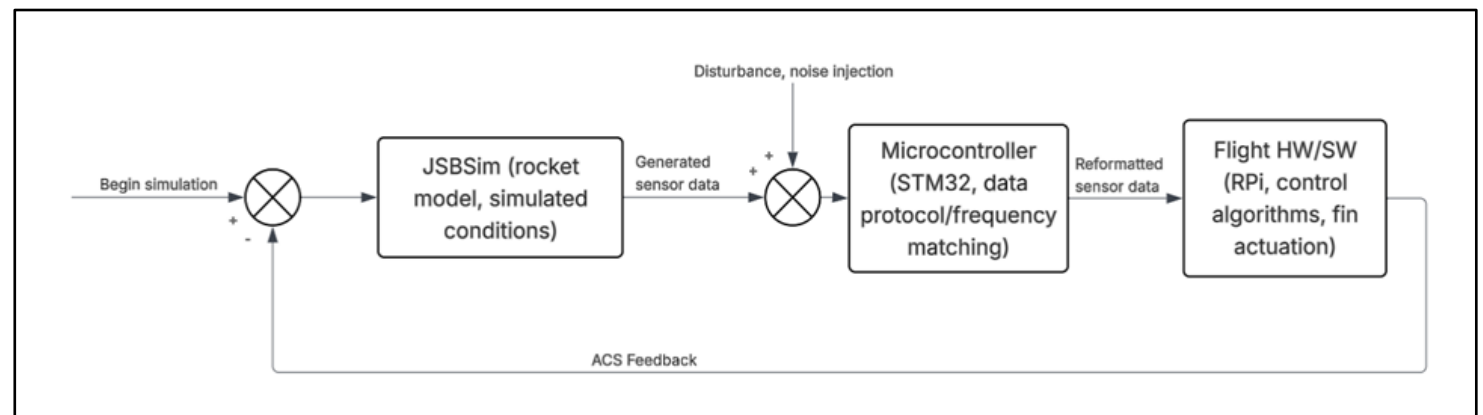


Figure 1: ACS Testing in Wind Tunnel

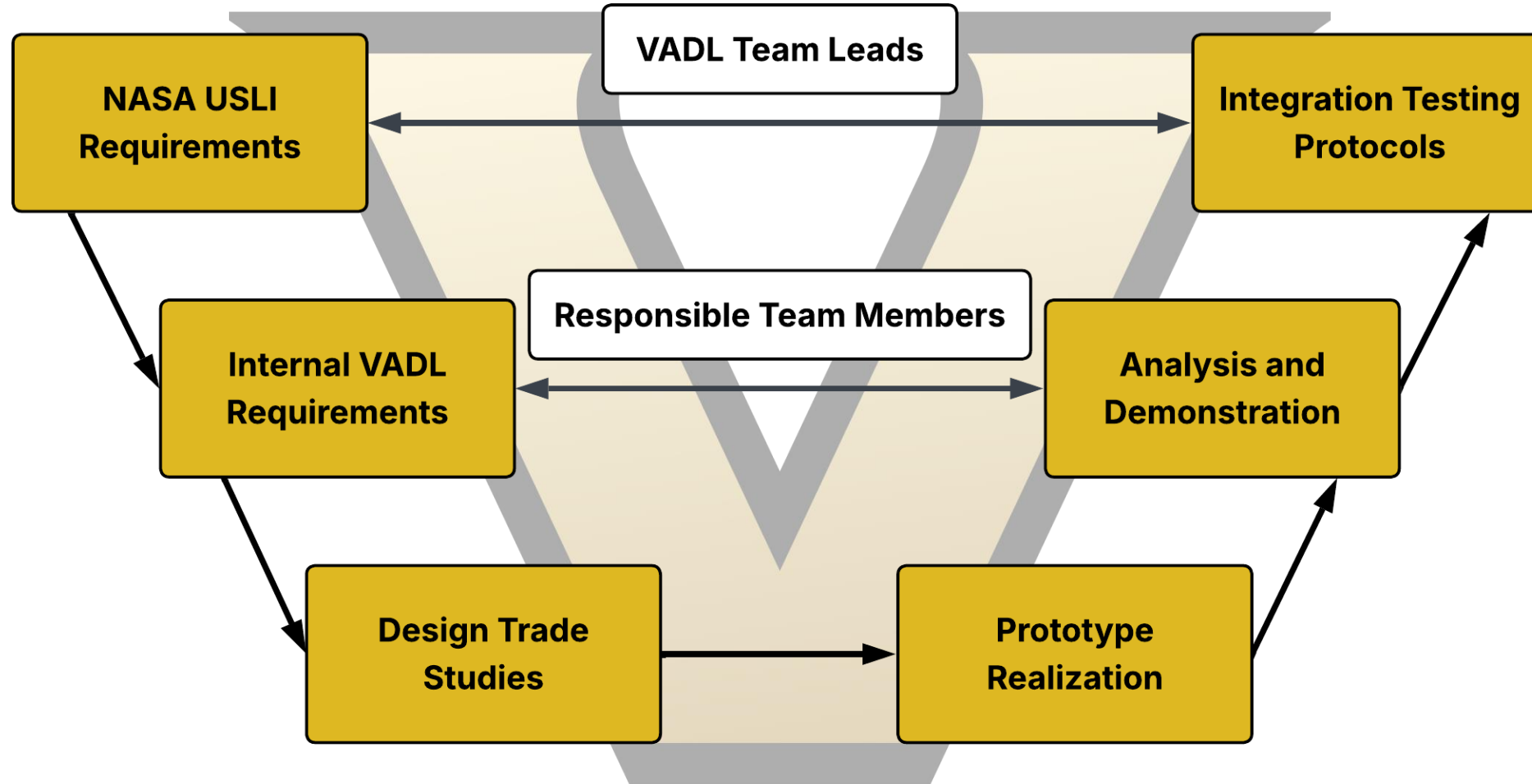
Figure 2: ACS Feedback Control Diagram



Requirements Compliance and Project Plan

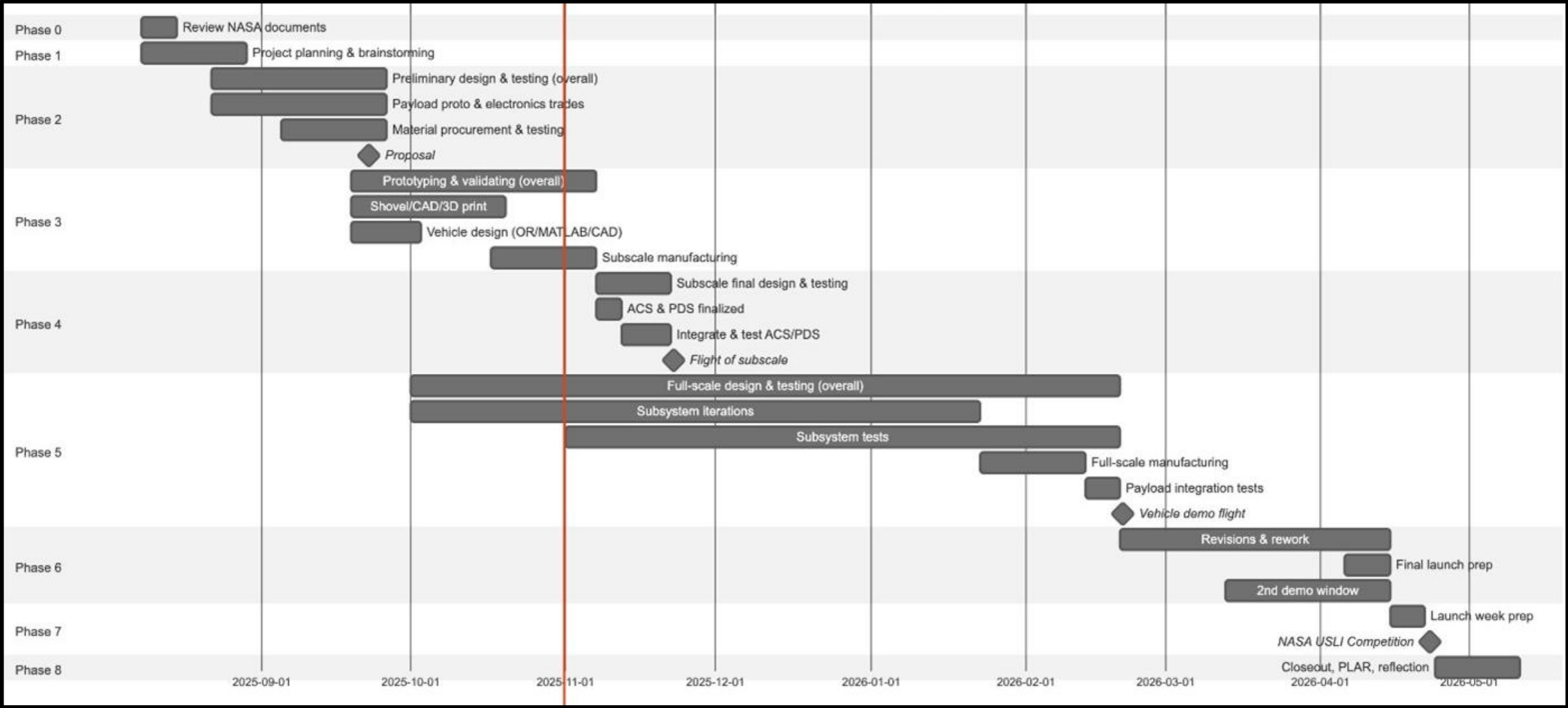


Requirements Verification





VADL Student Launch Timeline 2025-2026





Project Budget

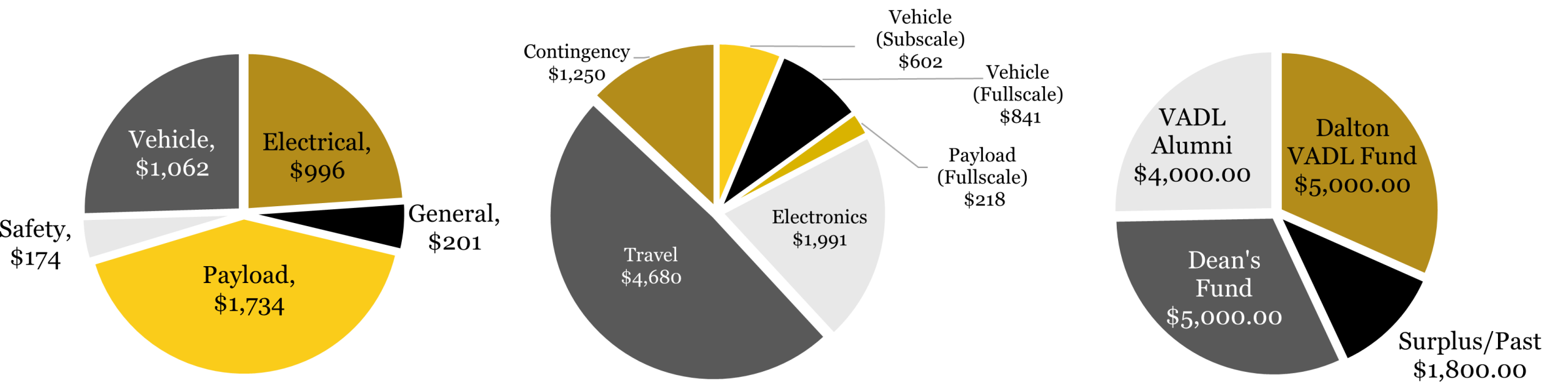
Current Expenses

+

Projected Expenses

<

Fundraising



Questions